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From data to diagnosis: a systematic review on AI-driven approaches to diabetes prediction

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Abstract

Diabetes mellitus is one of the most pressing global health challenges, and early prediction is critical to reducing complications, mortality, and healthcare costs. Conventional diagnostic tools remain limited, as they often rely on a small set of biomarkers and fail to capture lifestyle, genetic, or environmental risk factors. This review systematically evaluates how artificial intelligence (AI), including machine learning (ML) and deep learning (DL), enhances diabetes prediction by integrating multimodal data and improving clinical interpretability. Following PRISMA 2020 guidelines, a systematic search was conducted across PubMed, Scopus, IEEE Xplore, Web of Science, and ACM Digital Library for studies published between 2010 and 2024. Inclusion criteria required AI-based diabetes prediction models with reported performance metrics. From 2134 records, 155 studies met the criteria and were synthesized. AI models consistently outperformed conventional approaches (60–75% accuracy), Ensemble methods such as Random Forests and XGBoost achieved accuracy of 85–90% and AUC-ROC values > 0.90 (Abnoosian in BMC Bioinf 24:373, 2023. https://doi.org/10.1186/s12859-023-05465-z; Chen and Guestrin in Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining, Association for Computing Machinery, 2016. https://doi.org/10.1145/2939672.2939785). DL architectures showed notable strengths in unstructured data: CNNs achieved AUC > 0.95 in retinal image analysis (Gulshan et al. in JAMA 316(22):2402–2410, 2016. https://doi.org/10.1001/jama.2016.17216; Kermany et al. in Cell 172(5):1122–1131, 2018. https://doi.org/10.1016/j.cell.2018.02.010), while LSTMs reached 85–90% accuracy in continuous glucose monitoring predictions (De Bois et al. in Prediction-coherent LSTM-based recurrent neural network for safer glucose predictions in diabetic people, arXiv:2009.03722, 2020; Bian et al. in PLoS ONE 19(9):e0310084, 2024. https://doi.org/10.1371/journal.pone.0310084). Emerging methods such as federated learning enable privacy-preserving cross-institutional collaboration with comparable accuracy (~88%), while explainable AI techniques (e.g., SHAP, LIME, attention mechanisms) enhance transparency and clinical trust. Real-world case studies demonstrate improvements in early detection, reduced hospitalizations, and increased patient engagement when AI models are integrated into EHRs or mobile health apps. This review contributes a novel

In this article, acknowledgement section has been changed and updated.

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synthesis by combining methodological insights with clinical applications. The central takeaway is that AI-driven diabetes prediction offers significant advantages over traditional methods, but challenges in data quality, generalizability, fairness, and regulatory compliance remain. Addressing these will be essential to ensure safe, equitable, and clinically meaningful adoption in healthcare practice.

Keywords Diabetes prediction · Artificial intelligence · Machine learning · Deep learning · Feature engineering · Model interpretability

1 Introduction

Diabetes Mellitus (DM) is a metabolic dysfunction characterized by chronically high blood sugar levels resulting from abnormalities in insulin secretion, the body's response to insulin, or both (Loscalzo et al. 2022). This is a significant threat to international public health, as the increasing cases of this disease are further aggravated by the immense financial burden it poses to healthcare costs (Bommer et al., 2018). According to data from the International Diabetes Federation (IDF) in 2019, an estimated 463 million adults between 20- to 79-year-old individuals worldwide suffered from diabetes, projected to soon escalate to approximately 700 million individuals by 2045 (International Diabetes Federation 2019). The complications of diabetes include cardiovascular disease, renal disease, neuropathies, retinopathies, as well as higher incidences of amputations in the limbs, which significantly lead to morbidity as well as mortality (Narayan 2016). Apart from posing significant stress to the healthcare infrastructure, diabetes also incurs a substantial economic burden, encompassing indirect healthcare costs related to hospitalization, medications, specialized healthcare, disability, loss of productivity, and mortality (Zheng et al. 2018). Furthermore, diabetes also requires accurate diagnosis, as inappropriate treatments lead to increased complications, as well as poor patient outcomes.

Today, diabetes detection is focused primarily on traditional metrics, including blood glucose level and estimated HbA1c values. While diabetes is cared for reactively, after symptoms have appeared, by looking at blood glucose values, for instance, diabetes is diagnosed after alterations in metabolism have taken place in a patient's body in such a manner as to render pre-diagnosis efforts futile in this respect because of the paradigms of this study, which also only allow for diabetes diagnosis after serious alterations in a patient's metabolism have taken place, making pre-diagnosis in this respect futile as well. The truth is, diabetes is caused by complex variables in a patient's lifestyle. Thus, what is needed is an accurate predictive model to look at people in advanced stages before symptoms appear in them for effective diabetes prevention, which predictive models, based on complex variables, computers using artificial intelligence algorithms, for instance, can accurately pick up hidden, complex linkages between variables responsible for this disease's development in a patient's body because of a vast, heterogeneous set of variables drawn from electronic health care files, sensor data, imaging, genomic, and even behavioral data, which computers using artificial intelligence algorithms, for instance, can accurately pick up hidden, complex linkages between variables responsible for this disease's development in a patient's body because of a vast, heterogeneous set of variables drawn from electronic health care files, sensor data, imaging, genomic, and even behavioral data, which computers using artifi-

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Recently, this work has progressed using multimodal, lightweight models and hybrid models. As an example, Ding et al. (2024) showed that including clinical notes along with laboratory results using a multimodal model increased early detection of T2DM, while Sravanthi et al. (2023) and Abnoosian et al. (2023) showed that hybrid machine learning models using multiple data types resulted in strong predictive performance and computational efficiency.

Contrary to the usual models designed for analysis by predefined methods of statistics, AI models have the potential to deal with massive, metric-intensive, dynamically changing sequences of data, taking into consideration shifts in patient data, thus enhancing predictive capabilities after each iteration.

This study explores the applications of AI in diabetes prediction and risk factor analysis, as well as evaluating the integration of ML algorithms and DL techniques to provide more accurate diagnoses by focusing on early detection. The paper essentially discusses the following aspects of diabetes prediction, namely:

- How can AI algorithms lead to more accurate results in diabetes prediction as compared to other methods of diagnosis?
- What are the different Machine Learning (ML) methods, and Deep Learning (DL) approaches, and how do Machine Learning/DL models help in combining different types of data, including clinical, behavioral, genetic, and others, for predictive modeling of diabetes risk?
- What challenges, issues, and constraints are most pertinent to implementing AI-based systems for diabetes detection, diagnosis, and control?

This paper offers a detailed overview of AI methods for diabetes prediction, emphasizing advancements in feature engineering, interpretability, evaluation, and real-world use. It also presents examples of AI integration in wearable devices for Continuous Glucose Monitoring (CGM), highlighting their role in practical diabetes management systems.

This systematic review adhered to PRISMA guidelines published in 2020 to promote transparency and reproducibility in this study. The systematic search of scientific databases was undertaken for studies published between 2010–2024, which focused upon using Artificial Intelligence, Machine Learning, and Deep Learning for diabetes prediction, as only scientific studies published between those years using those methods would be relevant for analysis in this review. Included in this review were scientific studies that focused upon adequate methodology, but those publications that were in any language other than English,

as well as review articles, Scottish publications, and those publications that did not provide adequate data, were excluded from this review. Out of more than two thousand scientific publications, this review included only 155 scientific publications in this analysis.

The uniqueness of this review is reflected in integrating approaches in an applications-centered manner. Contrary to previous reviews, which either targeted machine learning approaches in detail or focused only on deep learning methods, this review extensively integrates classical and novel approaches, encompassing data types such as EHRs, imaging, wearable, and genomic data. A novel aspect of this review is that it provides a comprehensive integration of approaches for feature engineering, which is profoundly underrepresented in most previous reviews but plays an imperative role in modeling processes in AI applications in diabetes prediction. Furthermore, another novel component of this review is providing a comprehensive exploration of federated learning applications in an in-depth, informative manner, focusing on approaches related to maintaining patient confidentiality, which is of immense significance in healthcare applications of AI models for diabetes prediction. Furthermore, this review also tries to relate advancements in AI applications to diabetes prediction in relation to XAI approaches, which also makes this review novel. Four major aspects of this review make it original. First, this review systematically classifies diabetes prediction using AI, including both ML algorithms and DL approaches, across multiple data types. Second, this review provides a comprehensive analysis of predictive performance results from different data sets to identify the pros and cons of different algorithms for diabetes prediction using AI. Third, this review also covers current research areas in diabetes prediction, including federated learning, which is essential for diabetes prediction in healthcare settings. Fourth, this review also links algorithm development to real-world applications in healthcare, which is an original aspect of this manuscript, as most current review articles end with insights into algorithm development for different conditions, including diabetes.

The remaining paper is organized as follows: Sect. 2 provides an overview of previous work in diabetes prediction models. Section 3 provides information on data sources and data preprocessing techniques. Section 4 gives information on feature engineering for risk classification. Section 5 explains ML and DL algorithms for predicting diabetes. Section 6 gives details on evaluation metrics used to compare different methods for diabetes prediction. Section 7 addresses case studies and real-world applications. Section 8 addresses challenges and limitations. Section 9 concludes the major findings and areas of future research.

2 Background

Diabetes mellitus (DM) causes persistent high blood glucose due to insulin secretion or sensitivity abnormalities. (American Diabetes Association 2020). There are two main types of diabetes: type 1 and type 2 diabetes. Type 1 diabetes begins when the immune system annihilates pancreatic beta cells so that there is no insulin (Atkinson et al., 2014). On the other hand, type 2 diabetes develops as a result of the body becoming insulin-resistant and producing less insulin owing to obesity, lifestyle, or genetics (Kahn et al. 2014; Wild et al. 2004). Both types result in serious future complications such as heart disease, kidney damage, or eye problems that require early detection and immediate treatment (Gupta et al. 2025). A recent bibliometric review by Kiran et al. (2025) highlighted an accelerating trend

toward explainability and multimodal data integration in AI-based diabetes research, reinforcing the need for comprehensive methodological syntheses such as the present review.

2.1 Conventional methods for diabetes diagnosis

Traditional methods of diabetes diagnosis employ biochemical markers such as Fasting Plasma Glucose (FPG), oral glucose tolerance test (OGTT), and glycated hemoglobin (HbA1c) levels (World Health Organization 2019; Sacks et al. 2011; Gupta et al., 2017). These approaches toward diabetes predictions are being widely used but are not without limitations. For example, FPG and OGTT are affected by inter-day variability, stress, and medications, and can produce false positives or negatives (Fujimoto et al. 2017). They most often quantify one aspect of metabolism in isolation and thereby truncate their whole appreciation of glucose metabolism as it unfolds over time. HbA1c is widely used to monitor glucose levels over the long term, but it is unable to detect early presentations of diabetes or markers predictive of high risk for the disease. Despite their widespread use, these methods do not discriminate with adequate specificity for the earliest presentations of diabetes.

2.2 Limitations of conventional prediction methods

Existing methods of diabetes risk prediction are based primarily on traditional factors, such as age, genetic bias, body mass index (BMI), and physical inactivity (American Diabetes Association 2019; Tian et al. 2024). While the traditional models can potentially capture some of the risk groups, their potential for capturing early indications of diabetes is considerably undermined. This issue can be due to excessive dependence on a subset of the variables (Kahn et al. 2014; Gupta et al. 2018). The vulnerability to diabetes is influenced by an intricate interaction of environmental, lifestyle, and genetic factors; thus, traditional models possess the limitation of not being able to successfully capture these various factors (Sørensen et al. 2022). Key variables such as diet, sleep habits, and exercise levels are typically neglected even though they have an enormous impact on the overall risk (Hemmingsen et al. 2011; Hu et al. 2001). As such, conventional models fall short in their inclusion of some of these key variables with significant roles.

2.3 AI and ML in diabetes prediction

In order to be of great utility to medicine, AI must be capable of generalizing to new clinical populations and learning new things, not merely acting according to learned patterns. This capacity, sometimes referred to as generalizability, is still a great limitation. The majority of existing AI models function sub-optimally when used outside populations included in their training sets. Models trained on data sets that are mostly Caucasian in origin test below average when presented with data based on Asian or African populations. These limitations can be overcome by approaches such as transfer learning and domain adaptation, which have been developed by researchers to boost the capacity of models to gain knowledge about different types of datasets. These approaches are not limited by any constraints but should be completely justified on varied sources of information. Finally, although there is an immense opportunity available in the medical domain, there is a challenge concerning strong generalization, which remains an open problem. Finally, scalability is also a primary

problem because there is a need to offer optimum performance to time-constrained medical applications, dealing with a huge amount of information. For this purpose, there is a need to invest in sufficient computing resources and fault-tolerant computing architectures. To address these drawbacks, techniques such as transfer learning and domain adaptation have been proposed to improve model generalization across heterogeneous healthcare datasets (Kumar et al., 2021)

If these models are not refined and monitored over time, their performance can decline when exposed to new datasets. Federated learning, which enables institutions to collaboratively train models without sharing sensitive data, has emerged as a promising solution to improve scalability while maintaining privacy (Xu et al. 2020). However, existing federated approaches still face major challenges related to data heterogeneity across institutions and communication overhead, which can limit their overall efficiency (Chen et al. 2019).

2.4 Challenges in AI-based diabetes prediction

The field of AI for diabetes prediction is evolving and holds promise for enhancing its use for earlier diagnosis and prediction, but its adoption faces several challenges for effective use and utilization at a larger scale.

The greatest challenge is achieving good performance of AI models across diverse populations and heterogeneous healthcare environments. To be adopted and trusted extensively, the models have to be demonstrated to possess strong generalization beyond the training data sets utilized. In reality, however, patients vary in terms of age, ethnicity, and comorbidities, which leads to variation in model performance across different populations. The integration of AI in health care demands changes in existing health infrastructures, training and education of health professionals, and compliance and compatibility with medical standards. AI applications based on vast amounts of individual health data confront challenges of privacy and equity preservation. There are ethical concerns with AI applications in health care in terms of patient ownership of information, consent mechanisms for information sharing, and algorithmic decision-making fairness. Transparency, fairness, and strict data privacy controls are important aspects in the creation of ethical AI-based healthcare solutions.

Figure 1 illustrates some of the major difficulties or challenges faced within AI-assisted diabetes prediction systems grouped under categories named Validation, Bias, Privacy, and lastly, Usability. It is vital to overcome these limitations to guarantee success and higher acceptance within the field of diabetes prediction systems. Validation involves making outcomes general to many individuals and ensuring quality data for training and validation processes. Fairness is guaranteed by ensuring fairness within algorithms and the use of representative data to avoid any form of bias. Privacy involves ensuring that privacy is guaranteed while handling personal information and observing any required laws and statutes related to this field. Based on this explanation, acceptability is said to include integrating AI technology into patients' electronic files. This is accompanied by training needed within this medical field to guarantee all individuals possess the skills to operate these systems effectively for successful use and utilization within diabetes prediction systems.

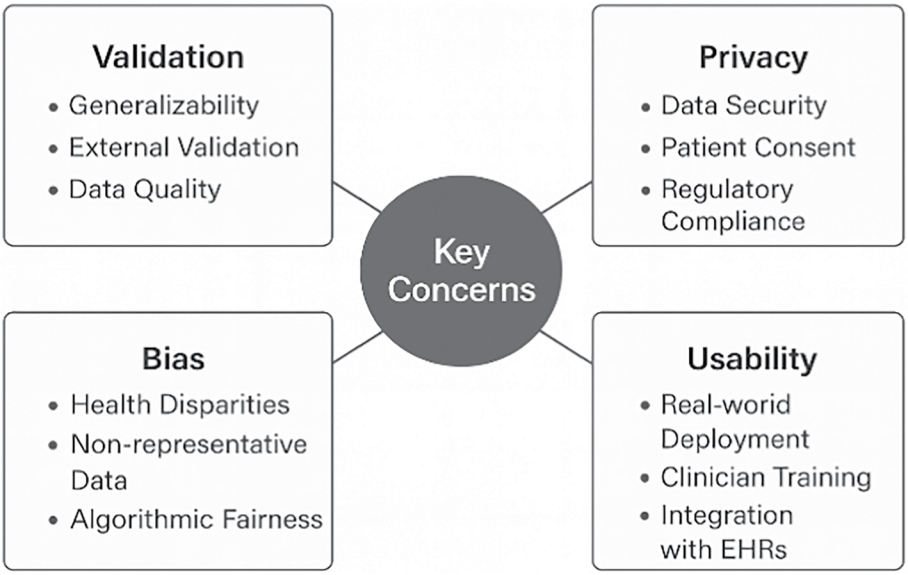

Fig. 1 Fishbone diagram illustrating key challenges in AI-based diabetes prediction, categorized into validation, bias, privacy, and usability

2.5 The future of AI in diabetes care

The future developments will witness major advancements being brought to diabetes care management through the use of AI. The role of AI is being widely utilized to revamp diabetes prevention and care programs for high-risk patients. Unlike before, these care programs do not use universal strategies but use each person's genetic information at once, while also focusing on lifestyle and medical aspects simultaneously, which is emphasized by Topol (2019). The effective outcomes of their research work demonstrate how such technology can potentially predict "the best possible strategies for treatment" and also determine "one's own risk to diabetes" very accurately. Additionally, the use of Artificial Intelligence-based technology is also capable of being connected to any kind of device that is present or offered to buyers within the market, such as glucose sensors and activity trackers, offering crucial information regarding one's bodily condition required for effective management of diabetes care (Ahmed et al. 2022; Heikenfeld et al. 2021).

However, usage of such technology is beset with a whole range of issues. These currently existent issues pertain to dependability within Artificial Intelligence-driven interventions, information protection and secrecy, and equality concerning accessibility to such technology across every social stratum. As developments take place, Artificial Intelligence possesses immense capability to boost forecasting accuracy and outcomes, making diabetes management much more efficacious.

3 Data sources and preprocessing

For diabetes prediction, high-quality datasets containing demographic, medical, and lifestyle information must be utilized. Another older approach is attribute classification into advantages and disadvantages, which are then evaluated against predefined criteria in relation to their effect on the model's predictive capability. The Pima Indians Diabetes Database (PIDD) at the UCI Machine Learning Repository includes 768 instances of 8 clinical attributes, such as the patient's age, the patient's BMI, and the patient's glucose levels (Smith et al. 1988). Although it has widespread use, primary immunodeficiency disease (PIDD) suffers from limitations, such as the availability of few samples and the study being conducted within a particular ethnic population, and therefore it is not generalizable to the population at large. The National Health and Nutrition Examination Survey (NHANES) database represents a huge database with health, diet intake, and lifestyle information that is non-specific for the population of America but rather is generalizable for the population of America (Centers for Disease Control and Prevention [CDC] 2018). The database serves as a valuable resource for exploring how diet and physical activity interact to influence diabetes risk across diverse populations. The NHANES data is collected on a self-report basis, which makes its quality heterogeneous, as well as subjecting its data to those that are inherent in the study design in a survey. A key resource for genetic, clinical, and lifestyle data is the UK Biobank, which has more than 500,000 participants (Sudlow et al. 2015). The dataset's scale and diversity enable large-scale analysis of environmental and genetic factors that affect diabetes risk. However, its complexity demands advanced analytical methods and substantial computing resources. The Framingham Heart Study is a long-term project that provides comprehensive information on cardiovascular disease and diabetes progression (Dawber et al. 2002). The present dataset is believed to provide relevant information on long-term diabetes progression and complementary factors. Yet, because of its distinct community, there could be constraints on making these findings more general. The four different datasets available contain differences in sample sizes, demographic variables, and other variables, which make them more suitable for distinct applications within diabetes development prediction models. They provide a collective basis to make an initial assessment and enhance distinct predictive models within different contexts. To make an optimal diabetes prediction model utilizing AI approaches requires a very precise identification of what dataset to select. A comparison table (Table 1) is shown to describe four distinct popular available datasets concerning sample sizes, significant variables, nature of both variables, advantages and disadvantages, and availability to conduct research studies.

As shown in Table 1, while PIDD is widely used as a benchmark, its limited size and single-population scope reduce generalizability. Conversely, NHANES and UK Biobank have more diverse data but are rather complex to preprocess, thus highlighting the complexities associated with accessibility as opposed to representativeness. It is notable from this comparison that while PIDD is currently widely regarded as a standard dataset because of its simplicity, it is lacking in terms of diversity to facilitate generalizability. Additionally, while UK Biobank and NHANES are rather complex to analyze, they contain diverse data to facilitate representativeness against simplicity.

Table 1 Comparative analysis of key datasets for diabetes prediction

Dataset	Sample Size	Key Features	Data Type	Strengths	Limitations	Data Accessibility
Pima Indians Diabetes Database (PIDD)	768	Age, BMI, Plasma Glucose, Insulin, Blood Pressure, Diabetes Pedigree Function	Tabular (Numerical, Categorical)	Standard benchmark dataset for diabetes prediction; well-structured with minimal missing data	Small dataset size; limited to the Pima Indian population, reducing generalizability	Public (Available via UCI Machine Learning Repository)
National Health and Nutrition Examination Survey (NHANES)	Varies (≈5,000 per cycle, nationally representative)	Demographics, Health History, Dietary Intake, Physical Activity, Lab Biomarkers (e.g., Fasting Glucose, HbA1c, Cholesterol)	Heterogeneous (Structured clinical data, survey responses, laboratory tests)	Large-scale, diverse U.S. population; includes lifestyle factors, biomarkers, and comorbidities	Self-reported data introduces recall and response biases; a periodic survey structure may lead to inconsistencies across cycles	Public (Available via CDC, some restricted access for sensitive data)
UK Biobank	500,000+	Genetic Variants (SNPs), Clinical Measurements, Lifestyle Factors, Biomarkers (e.g., HbA1c, Glucose, Lipid Panels), Imaging Data	Multi-modal (Genomic, Clinical, Behavioral, Imaging)	An extensive dataset with deep phenotyping and genetic data, allowing integration of polygenic risk scores	High computational complexity for preprocessing; potential selection bias due to voluntary participation	Restricted (Requires application and approval, fee-based access)
Framingham Heart Study (FHS)	≈14,000 participants (Three generations, ongoing cohort)	Cardiovascular & Metabolic Risk Factors (e.g., BMI, Blood Pressure, Lipid Profiles, Diabetes Incidence)	Longitudinal (Time-series, Repeated Measures)	Long-term follow-up enables tracking disease progression; useful for predictive modeling with temporal dynamics	Primarily Caucasian cohort; findings may not generalize to ethnically diverse populations	Restricted (Access granted via application and data use agreements)

3.1 Feature selection and data preprocessing techniques

To make precise estimates of diabetes, models consider a complex variety of data inputs: personal variables (gender and age); physiological variables such as BMI and waist-to-hip ratio; clinical markers such as fasting glucose and HBA1C; and lifestyle variables such as physical inactivity, diet, and smoking status. However, all these factors are commonly recognized as significant risk factors for type 2 diabetes. Feature selection is very important for model interpretability and accuracy. It is a complex optimization process consisting of identifying the most relevant input features for classification, eliminating redundant and irrelevant data, and ensuring the uniqueness of each parameter. Some methods, such as recursive feature elimination and mutual information, are applied to tune these features.

3.2 AI-based advances in diabetes prediction

Research has inevitably proved that genetics are the determinants of the progression toward diabetes (Thompson et al., 2024). It was originally proved by Thompson et al. (2018) and later proved by Gan et al. (2019). The combination of genomics with other health sources of data improves predictability and allows for more accurate estimation of the risk. Additional techniques, such as Multiplication by Chained Equations (MICE) with the traditional techniques for the handling of the missing data, are more accurate compared to the traditional techniques, as discussed by van Buuren (2018), Stekhoven and Bühlmann (2012) and Liu et al., (2021).

Moreover, machine models such as Isolation Forest and DBSCAN (Density-Based Spatial Clustering of Applications with Noise) can cleanse the precision of prediction by eliminating the latter since Liu et al. (2008), Ester et al., (1996) and Xu et al. (2015) proved it so. The latter elimination is important for the prediction of diabetes since model far-outliers are likely to impair model functionality. With these new models, precision, validity, and generalizability for the prediction of diabetes are cleansed for varying populations and situations. CGM and wearable signals often display irregular and partially coherent dynamics; predictive models must be stress-tested for robustness. Insights from chaos–coherence analysis show how coherence influences parameter stability in complex systems, offering a framework for testing whether small perturbations in physiological signals shift risk predictions ( ). While preprocessing ensures the dataset is clean and consistent, feature engineering builds on this foundation by identifying and constructing variables that maximize predictive power. Section 4 discusses these approaches in detail.

3.3 Systematic review protocol (PRISMA 2020)

This systematic review adhered to guidelines outlined for Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA 2020).

3.3.1 Databases and search strategy

A comprehensive literature search was conducted for peer-reviewed articles published between January 2010 and December 2024 on PubMed, Scopus, IEEE Xplore, Web of Science, and ACM Digital Library databases. The final search was conducted on 31 December

2024. Search terms were combined using controlled vocabulary terms and free-text terms, for example:

("diabetes" OR "type 2 diabetes" OR "diabetic prediction") AND ("artificial intelligence" OR "machine learning" OR "deep learning" OR "neural network" OR "predictive model").

3.3.2 Inclusion criteria

1. Studies applying AI, ML, or DL techniques to predict diabetes risk, onset, or diagnosis.
2. Articles reporting quantitative performance metrics (e.g., accuracy, AUC, precision, recall).
3. Peer-reviewed, English-language publications.

3.3.3 Exclusion criteria

- Review papers, commentaries, editorials, or conference abstracts without full text.
- Non-English publications.
- Studies lacking methodological transparency or performance data.

3.3.4 Screening process

A total of 2,134 records were retrieved, and after removing duplicates, title and abstract screening were conducted by two independent reviewers. A total of 159 studies qualified for full-text analysis based on all criteria for inclusion. Any discordance was resolved through consensus between reviewers or by a third reviewer if needed.

3.3.5 Data extraction and quality assurance

The extracted data consisted of the year of publications, dataset information, modeling techniques, feature engineering methodology, and the measures of evaluation. This extraction was done using a standardized template and verification processes. The method for writing this systematic review adheres to PRISMA 2020 guidelines for reproducibility and bias reduction.

4 Feature engineering

The most important step involved in predicting diabetes using artificial intelligence is feature engineering. The feature engineering phase comprises processes such as choosing, shaping, and constructing relevant features to identify trends and make predictions about diabetes with considerable accuracy. The feature engineering process comprises processes such as choosing, shaping, and constructing variables. The features identify relevant patterns that help make predictions. Removing unwanted features, balancing features, and extracting relevant features make it an optimal predictor. The main tasks involved in feature engineering

to make predictions about diabetes are gathering information, preprocessing information, and extracting features. The processes involved in extracting features comprise Recursive Feature Elimination (RFE), Lasso, PCA, and t-SNE. This is shown in Fig. 2.

4.1 Techniques for feature selection

Feature Selection is a method to identify important variables within a database with a minimum number of redundant variables. Feature Selection improves model quality, avoids over-training, and helps to access more information about variables. For diabetic prediction, each method has pros and cons, which play an important role in the choice of a correct method. The correlation method attempts to find a link between the prognosis result of diabetes and its features. The features are strongly correlated within FPG and within HbA1c, which is difficult because of the presence of multicollinearity within the outliers. This is resolved using Recursive Feature Elimination, which removes irrelevant features to identify important features. RFE is effective with high-dimensional data, making it valid in models with complex data for predicting diabetes. Guyon et al. (2002) confirmed that RFE enhances the quality of SVM-based models by discarding irrelevant features. The inclusion of a penalty term in L1 regularization, known as Lasso, forces irrelevant features to become zero. Ashiquzzaman et al. (2017) conducted similar research in selecting key factors such as BMI, genetics, and age that enhance models' efficiency and interpretation. Mutual information shows the relationship between the target variable and its features, but this can be complicated. Wang et al. (2018) utilized the strategy in detecting genetic and lifestyle variables, i.e., body activity and dietary intake, that enhance the predictiveness of the model. The

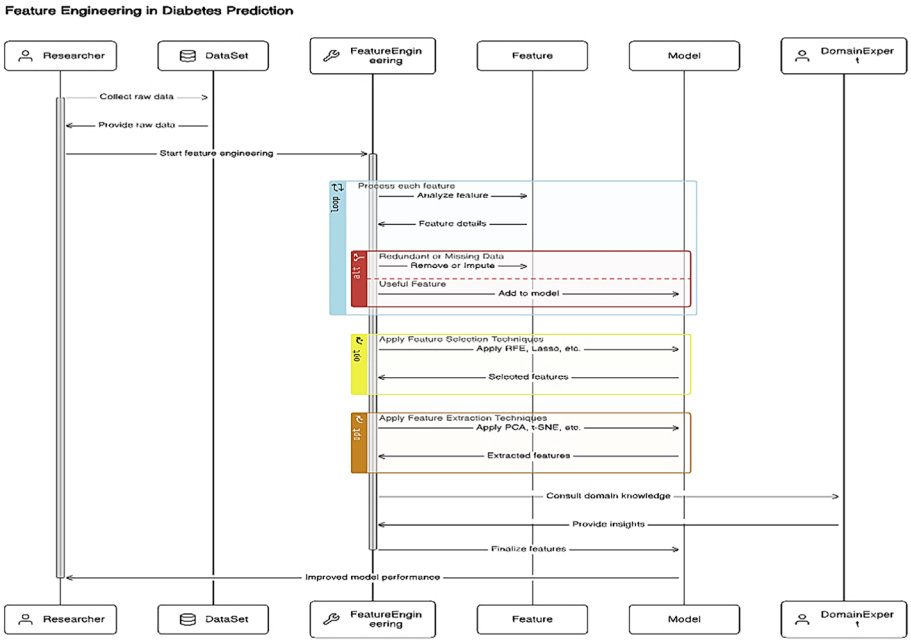

Fig. 2 Feature engineering workflow for AI-driven diabetes prediction: from raw data to optimized model performance

strategy of a Chi-square test discovers target variable dependency on categorical variables and discards variables with high dependency. Sadeghi et al. (2022) used this tool to predict a patient's genetic and smoking profiles. By judiciously employing these methods, they hope to enhance the interpretability of large-scale diabetes models. Table 2 compares five methods on algorithmic type, scalability, computational complexity, and noise resistance.

As demonstrated in Table 2, techniques such as Recursive Feature Elimination (RFE) or Lasso should be applied to high-dimensional examples, as are prevalent in biomedicine and genetics analysis, to make these outcomes interpretable and avoid risks of overfitting associated with classification approaches. Simpler techniques may include correlation analysis

Table 2 Comparative overview of feature selection methods for diabetes prediction

Method	Algorithm type	Scalability	Interpretability	Computational complexity	Robustness to noise	Use cases in diabetes prediction
Correlation Analysis	Univariate	High	High	Low	Low	Identifying linear relationships between features like fasting plasma glucose and diabetes risk
Recursive Feature Elimination (RFE)	Wrapper	Moderate	Moderate	High	Moderate	Feature selection for high-dimensional datasets improves model efficiency by eliminating non-significant variables
L1 Regularization (Lasso)	Penalty	Moderate	High	Moderate	Moderate	Used for high-dimensional data like genetic markers, eliminating irrelevant features for more parsimonious models
Mutual Information	Filter	Low	Moderate	High	High	Handling mixed data types in complex datasets, like combining clinical and lifestyle factors to predict diabetes risk
Chi-Square Test	Statistical	High	High	Low	Moderate	Selection of categorical features (e.g., smoking status, family history) based on statistical dependence on the target variable

and may result in interpretable outcomes, but lack information on non-linear relationships between features and classes. This shows how important it is to tailor approaches to both features and dataset type. Feature extraction techniques directly relate to results, both concerning accuracy and interpretability. Although PCA and autoencoders remove redundancy and identify hidden aspects within a dataset, they lack transparency.

4.2 Feature extraction methods

Feature extraction is a method to address high-dimensional data more effectively, resulting in reduced computation time. PCA is another method for dimensionality reduction to pick an axis with maximum variance to preserve more informative features. (Zhu et al. 2019) analyzed how PCA can increase diabetes prediction using a logistic regression model by removing noise features and overcoming multicollinearity due to PCA. The disadvantages of PCA are that it is restricted to linearity and requires variance measurement solely to prove significance. PCA can lead to misinterpretation while dealing with non-linear features, where t-distributed Stochastic Neighborhood Embedding (t-SNE) is equally proficient, with a unique capability to plot high-dimensional features. In the context of diabetes research, Meniailov and Krivtsov (2022) applied t-SNE for dimensionality reduction of patient data, demonstrating how lower-dimensional embeddings can aid downstream classification or visualization tasks. Autoencoders are neural network-based techniques that can compress the data effectively without losing significant features. Autoencoders are composed of an encoder for reducing the dimensionality and a decoder for the reconstruction of the original data. Folorunsho et al. (2024) illustrated that autoencoders are better than conventional approaches in analyzing intricate datasets, particularly in identifying gene-associated and clinic-associated component clusters for better predictive tasks.

Some of these feature extraction methodologies, such as PCA, t-SNE, autoencoders, and ICA, present different advantages based on the kind of data. Table 3 compares methods according to dimensionality reduction, complexity handling, variance retention, and suitability for non-linear data. The comparison shows that the techniques perform well and provide substantial support to researchers to overcome current model limitations, especially for complex clinical and genetic data.

4.3 Role of domain knowledge

Experts apply universal principles in assessing diabetes risk factors. BMI, blood glucose, and family history rank as significant predictors of diabetes. Ren et al. (2018) highlight that incorporating expert knowledge into modeling not only simplifies the process but also significantly enhances model accuracy. The era of relying solely on traditional indicators is coming to an end. Incorporating broader datasets, such as complete patient histories, genetic information, and lifestyle factors, greatly enhances the precision of predictive algorithms. For instance, Hodgson et.al (2022) demonstrated that integrating genetic variables with traditional measures like HbA1c and blood pressure significantly improves predictive accuracy in assessing individual diabetes risk. Missing or incomplete data is another area where domain expertise plays a crucial role. As Hyvärinen and Oja (2000) note, handling missing data is not a simple “fill-in-the-blanks” task; rather, it requires careful consideration of context-specific approaches, such as whether or not to impute values, each carrying its

Table 3 Comparative overview of feature extraction techniques for diabetes prediction

Method	Dimensionality reduction	Computational complexity	Preservation of variance	Non-linearity handling	Data type applicability	Use cases in diabetes prediction
PCA	Linear		High	No	Continuous	Reducing the dimensionality of clinical datasets, removing multicollinearity, and improving logistic regression model performance
t-SNE	Non-Linear	High	Moderate	Yes	Continuous & Categorical	Visualizing high-dimensional diabetic datasets to uncover patient clusters and trends
Autoencoders	Non-Linear	High	High	Yes	Continuous	Feature extraction from complex datasets (e.g., electronic health records) for DL-based diabetes prediction models
ICA	Non-Linear	High	Moderate	Yes	Mixed Data	Extracting independent components from clinical, genetic, and lifestyle data to enhance prediction accuracy in diabetes models

own level of uncertainty in data analysis. Temporal data further complicates this issue, as data varies from time to time, as explained in detail in (Rajkomar et al. 2018). Essentially, domain expertise is quite valuable in this respect, as it makes sure that predictive models have adequate data to work upon in both statistic as well as real-world opinion, which thus makes them possible to work as far as real-world applications go by having full validity in them, as indicated in Fig. 3 below, in which Random Forest modeling rates variables in proportion of priority to diabetes, signifying most significant variables in this respect in detail.

4.4 Impact on model accuracy

Feature engineering is among the most powerful ways to enhance predictive models' efficiency. Feature engineering helps overcome overfitting, while also making models more

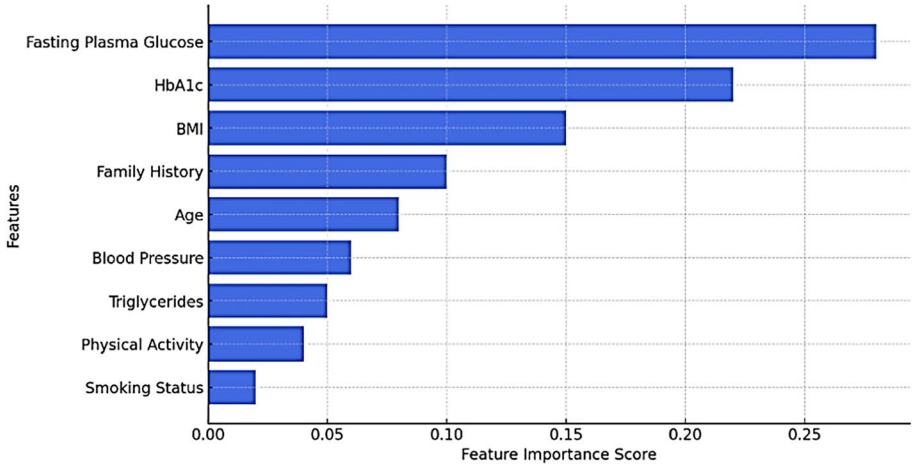

Fig. 3 Feature importance ranking for diabetes prediction using a Random Forest model

interpretable and responsive to new data. For instance, Kalusivalingam et al. (2022) explained that an accurate selection of features led to a relative improvement of 15% in their Random Forest classification model's accuracy. Appropriate selection of relevant features plays a crucial role in the validity of the model itself. According to Tibshirani (1996), the optimal results for recall and precision were achieved in feature selection through the use of the LASSO technique. The usage of dimensionality reduction algorithms, PCA and Autoencoders, becomes priceless in this context.

A combination of logistic regression and PCA, as suggested by (Huang, Li, & Zhang, 2020), resulted in an approximately 10% improvement in accuracy. In a similar context, the study by Gupta et al. (2022) found that the performance of autoencoders outperformed conventional methods in unearthing valuable knowledge from EHRs. In the context of the above-mentioned research works collectively, the importance of "Feature engineering" in developing an efficient model from a medicinal point of view cannot be overlooked.

4.5 Challenges in feature engineering

Feature engineering is another essential technique that enables predictive models to extract new, meaningful insights while reducing the risk of overfitting. It also enhances the interpretability of these models, allowing their outcomes to be more transparent and understandable. As discussed by Omranian et al. (2023), it has been established that by applying feature engineering, the predictive accuracy of a Random Forest classification model increased by up to 15%. The relevance of features is an important consideration in selecting features for predictive modeling, as it may lead to building an even more accurate predictive model than in this study, as suggested by Gupta et al. (2022). A similar study showed that Autoencoders are much better than traditional approaches for obtaining significant information from EHRs.

4.6 Future directions in feature engineering

Feature engineering remains the basis of precise and interpretable AI-driven diabetes prediction models. Feature selection methods (RFE, Lasso regression) help overcome redundancy and improve generalizability, while feature extraction methods (PCA, t-SNE, and autoencoders) identify hidden patterns in diverse multimodal data sets. The addition of professional clinical judgment helps ensure that the engineering of new features remains grounded in clinical realities and statistical significance. Feature engineering in AI diabetes models will continue to work toward improving model accuracy and interpretability through the bridging of the divide between raw data sets and valid clinical knowledge. This will continue to remain relevant as the field of AI-driven diabetes research shifts toward automated and interpretable feature learning.

In short, the car industry will be poised to assume even greater control of feature engineering through the utilization of the capabilities of AutoML and advanced algorithms of feature learning, which until now are even more dependent on the domain specialist. Simultaneously, biological data, especially in the realms of genomics, metabolomics, and proteomics, will be poised to assume an even more crucial role in the further development of diabetes risk models through the utilization of the vastly deeper biological viewpoint. Finally, the incorporation of AI will be poised to assume a crucial role in the further development of AI models of healthcare through the utilization of AI transparency.

In conclusion, feature engineering is thus the cornerstone for an accurate and interpretable AI-based diabetes forecasting system. Feature selection methods (RFE, Lasso) help in eliminating redundancy in features, making them more generalizable, whereas methods like PCA, t-SNE, and autoencoders help in extracting hidden patterns in complex multimodal datasets. Application of domain knowledge is thus essential to ensure that features engineered for prediction purposes in diabetes forecasting systems remain biologically as well as statistically accurate. However, collectively, all methods work in concert to provide enhanced precision, preventing cases of over-fitting, as well as increasing interpretability, which leads to filling up the gap between data/information as well as concrete biological insights, respectively.

5 AI techniques for diabetes prediction

In recent years, AI has greatly evolved the process of diabetes prediction in the healthcare industry. Conventional methods of diabetes prediction using statistics in healthcare settings have been believed outdated by the recent advancements of AI, which have accessed complex patterns in large sets of data in an efficient manner that is more accurate than conventional methods (Goodfellow et al. 2016; Chollet 2018). Presently, predictive analytics for diabetes by combining patient data and large population data has enabled AI to create an accurate predictive model for healthcare practitioners to assess patient risks more accurately (Esteva et al. 2019). This review presents a comprehensive analysis of various AI models applied to diabetes prediction, ranging from traditional machine learning approaches to advanced deep learning architectures.

The selection of different AI approaches in this review is informed by their appropriateness for different data types and diabetes prediction scenarios. Conventional machine learn-

ing approaches like logistic regression, decision trees, and SVMs were selected because of their efficacy in interpretable, structured tabular data (eHRs, patient demographics). Ensemble learning approaches like Random Forest, XGBoost, and LightGBM were selected because of their capacity to handle overfitting, increase robustness, and perform well on diverse data types. Deep learning approaches, including convolutional, LSTM, GRUs, and autoencoders, were informed by their capacity to efficiently extract complex features from data, including convolutional networks for extracting features in hierarchies in retinal scans, LSTM-GRUs for modeling temporal dependencies in cG, and in eHR sequences, autoencoders for automatic feature learning, as well as modeling complex hierarchies in cG sequences. It is worth noting that recent state-of-the-art models, such as lightweight transformer architectures (Wu et al. 2022) and multi-task learning frameworks (Tao et al. 2025), represent the latest advancements in improving efficiency, scalability, and performance in diabetes prediction. Transformers, as a recent innovation in model sequencing, incorporate novel attention views for better modeling of long dependencies in sequencing data, such as cG sequencing data. Finally, more advanced approaches, including federated learning, allow for synchronized, shared model training without central data repositories, meaning data is never shared, which is beneficial for data protection, patient data protection regulations, as well as insights into more interpretable AI (XAI) approaches, including SHAP, LIME, attention views, providing an emphasis for better interpretability, which is paramount for increased usability in diabetes predictive modeling applications.

To help applied researchers and clinicians quickly identify which AI approaches are most suitable for different data sources, Fig. 4 maps the relationship between data types and corresponding model families. This mapping emphasizes that model selection should depend on the structure and modality of the input data, ensuring that the chosen algorithms align with clinical data characteristics and computational feasibility.

Figure 5 below provides an integrated concept map of common machine learning algorithms, focusing on significant data sets such as electronic health records and output from wearable devices, that provide vital inputs to algorithms. While basic algorithms allow for early signs of diagnosis, more opportunities lie in deep learning algorithms in terms of unlocking complex, novel patterns, whereas ensemble learning algorithms allow for better

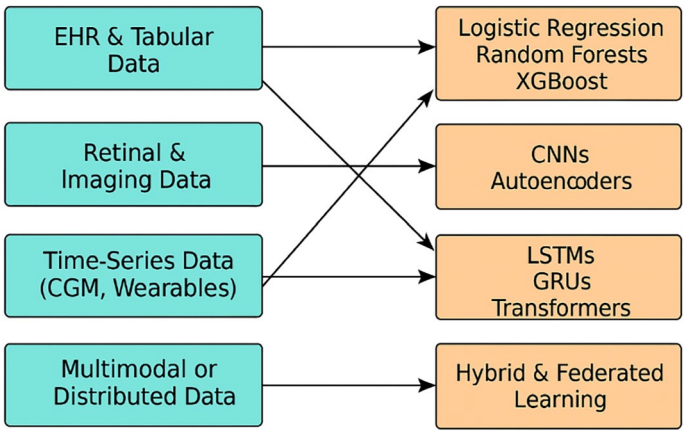

Fig. 4 Mapping of data types to suitable AI models for diabetes prediction

predictive analytics by leveraging diverse expertise in different algorithms, expert systems helping to make them more interpretable.

5.1 Base machine learning models

The conventional algorithms, such as decision trees, support vector machine (SVM) algorithms, k-nearest neighbors (KNN) algorithms, and logistic regression algorithms, have played an important role in identifying diabetes risk factors and in estimating the chances of diabetes development. These algorithms work efficiently in identifying basic patterns in patient data, which include BMI values, blood glucose levels, and other variables fundamental to creating an efficient predictive model for diabetes development. The basic algorithms shall also be discussed in detail in the succeeding chapter, together with their roles in the diabetes prediction process.

5.1.1 Logistic regression

Logistic Regression is a binary data classification statistical method and thus very suitable for diabetes prediction based on clinical attributes like BMI, age, and blood glucose. Logistic function predicts the probability of the binary outcome based on real-value input scaled between 0 and 1. The convenience and interpretability of logistic regression have made it a favorite in clinical environments where these attributes are useful in the application of evidence-based medicine to guide clinical decision-making (Hosmer et al. 2013; Liang and Zeger 1986).

Technically, logistic regression applies the maximum likelihood estimation (MLE) method to calibrate the model parameters such that the model is composed to make the optimal possible predictions of diabetes risk. Logistic regression does assume linearity in the relationship between the predictors and the log-odds of the response variable and thus has a

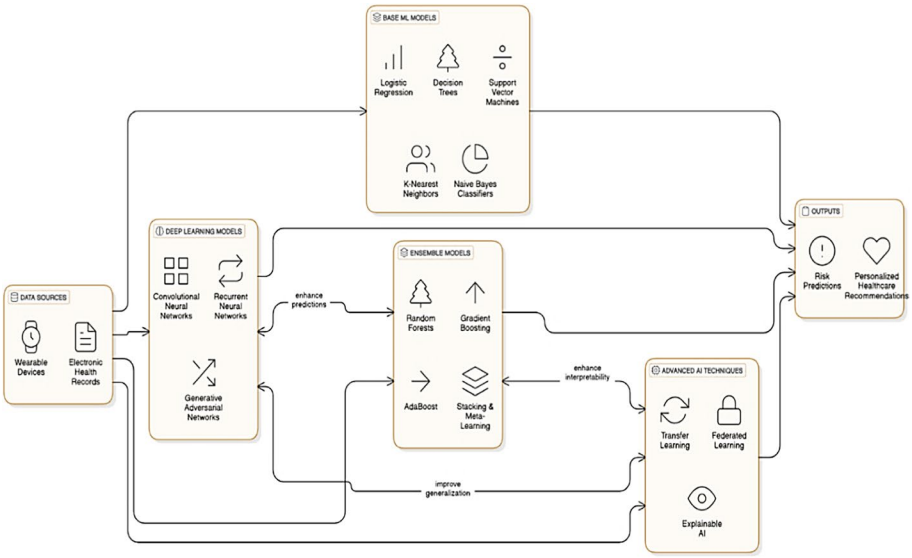

Fig. 5 Overview of ML algorithms for enhanced diabetes prediction

poor capacity for detecting non-linear relationships (Pindyck and Rubinfeld 2013). To counteract this weakness, regularization methods, i.e., L1 (Lasso) and L2 (Ridge), have been used to avoid overfitting and enhance model performance in scenarios of high-dimensional data (Tibshirani 1996; Zou and Hastie 2005). These methods work by introducing penalties on excessively large coefficients to favor generalization, particularly in diabetes datasets that are typified by pervasive multicollinearity (Salehi et al. 2019). Recent studies have further explored the use of logistic regression with advanced regularization techniques for diabetes prediction, achieving improved performance in imbalanced datasets (Muningsih et al. 2024). Recent work continues to emphasize interpretable, population-level modeling; for instance, Li et al. (2025) applied explainable ML to nationwide data from the China Health and Nutrition Survey, achieving robust pre-diabetes prediction while maintaining interpretability.

5.1.2 Decision trees

Decision Trees recursively divide the data at attribute thresholds to produce an interpretable model. Decision tests such as blood pressure, glucose, or BMI have a tree-like structure with decision nodes. The decision path interprets the model and forecasts diabetes risk. Decision trees are simple to operate but are vulnerable to overfitting in complex models (Breiman et al. 1984). Pruning algorithms like cost-complexity pruning remove weak predictive nodes (Loh 2014). Node partitions are executed utilizing metrics, including entropy and Gini impurity, to optimize information gain. Through feature selection and ensemble methodologies, decision trees have proven to be the most effective for diabetes prediction (Liaw and Wiener 2002). Visual aids facilitate clinicians' comprehension of decision pathways, thereby increasing their confidence in the model (Sen Kaya et al. 2023). Contemporary developments, such as hybrid pruning, have enhanced the predictive performance regarding diabetes (Talari et al. 2024).

5.1.3 Support Vector Machines (SVMs)

SVMs are suitable for high-dimensional spaces and are used in diabetes prediction as datasets contain many features such as glucose levels, BMI, and blood pressure. SVMs select the optimal hyperplane maximizing class margins for better generalization (Cortes and Vapnik 1995; Guyon et al. 2002). The main advantage of SVMs for diabetes prediction is that they can handle non-linear decision boundaries by using kernel functions such as RBF and polynomial kernels (Vapnik 1998) to map feature spaces for better classification of diabetes risk. However, they are limited by large-scale parameter tuning, usually through grid search or Bayesian optimization, and computing power in big data (Chang and Lin 2011).

5.1.4 K-nearest neighbors (KNN)

K-nearest neighbors (KNN) is a non-parametric classifier that assigns labels based on the majority vote of the k-nearest neighbors. KNN is very convenient for diabetes risk classification, especially in datasets with clearly defined clusters of diabetic and non-diabetic individuals. One of the major drawbacks of KNN is computational complexity, especially when dealing with large datasets. To overcome this problem, KD-trees and Ball-trees are

used to accelerate nearest-neighbor search (Cover and Hart 1967). Distance-weighted KNN gives more weight to near neighbors and improves the accuracy of diabetes prediction (Jolliffe 2002). KNN variants with adapted distance metrics or sampling corrections are commonly explored in imbalanced ML tasks, though definitive superiority in diabetes prediction remains to be established (Salmi et al. 2024).

5.1.5 Naive Bayes classifiers

Based on Bayes' theorem and the assumption of feature independence, naive Bayes is an approximation and is effective on large data sets. Gaussian naive Bayes uses a normal distribution for numeric attributes and is suitable for continuous clinical measures like blood pressure and blood glucose levels (Domingos 2000). However effective, naive Bayes is sensitive to violations of the independence assumption. Since diabetes datasets have conditional independence, naive Bayes is an appropriate classifier for classification (Faruque et al. 2019). Laplace smoothing improves performance to handle zero probabilities in categorical attributes.

5.2 Deep learning models

DL models predict diabetes by exploring nonlinear relationships in large datasets efficiently. Different architectures such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, Gated Recurrent Units (GRUs), Deep Neural Networks (DNNs), autoencoders, Restricted Boltzmann Machines (RBMs), and transformer models have been explored for diabetes risk assessment in early diagnosis.

5.2.1 Convolutional neural networks (CNNs)

CNNs are widely adopted in medical image analysis for the detection of diabetic retinopathy in fundus images (Gulshan et al. 2016). CNN-based architectures consistently achieve AUC > 0.95 for retinal image analysis tasks (Gulshan et al. 2016; Kermany et al. 2018), demonstrating superior capacity to detect spatial patterns in fundus photographs compared with traditional ML methods. CNNs with convolutional and pooling layers learn spatial features automatically without manual preprocessing of data. CNN models for diabetic retinal irregularity combine ResNet, VGG, and Inception classification features (Kermany et al. 2018). CNNs analyze glucose trends and physiological sensor data and show hierarchical feature representation (Yu et al. 2019). Predicting diabetes is challenging due to the requirement for vast labeled data for CNNs. CNN-based models suffer from temporal processing, making them unsuitable for sequential patient health records. Advances in CNN architecture, such as attention mechanisms, have enhanced diabetes prediction recently (Qi et al. 2024).

5.2.2 Recurrent neural networks (RNNs) & LSTMs

RNNs are useful for predicting diabetes from CGM and EHR data by learning time series dependency in data (Xiao et al. 2018). LSTM and GRU models trained on CGM time-

series typically achieve 85–90% accuracy (De Bois et al. 2020; Bian et al. 2024), capturing temporal dependencies essential for predicting glucose fluctuations. The vanishing gradient problem in RNNs hinders long-term dependency learning in data. Memory cells and gated architectures for storing long-term dependencies improve diabetes prediction in LSTM networks, an advanced RNN (De Bois et al. 2020). Glucose readings were predicted using historical glucose, insulin injections, and food intake using LSTMs (Bian et al. 2024). Building on this, coherence-aware methodologies have been proposed to capture variability indices and intermittency features that reflect nonlinear physiological fluctuations. Recent chaotification and multipartite correlation frameworks highlight how such features can enrich sequential prediction models and improve robustness in hybrid healthcare systems (Bary et al. 2023). LSTMs have high computational time and overfitting risks with small datasets. LSTMs are complex for clinicians and unclear in model decisions. Recent studies propose lightweight LSTM variants for diabetes prediction with reduced computational overhead while maintaining accuracy (Liu et al. 2023).

5.2.3 Gated recurrent units (GRUs)

GRUs are a good fit for sequential diabetes prediction models due to their reset and update gates, which help regulate information flow and alleviate sparsity in patient history data (Cho et al. 2014). GRUs have state-of-the-art performance in blood glucose and diabetes onset prediction using fewer parameters than state-of-the-art RNNs (Koca et al. 2024). GRUs are more interpretable than LSTMs but sensitive to hyperparameters and need to be optimally tuned. Clinical sequential data usually has poor quality. Recent research investigates GRUs with attention mechanisms for better diabetes prediction (Barbato et al. 2025).

5.2.4 Deep neural networks (DNNs)

DNNs with multiple hidden layers are the most utilized DL model for diabetes classification. DNNs skip shallow ML methods by taking into consideration factors, including lifestyle and clinical data, and are more accurate (Shickel et al. 2018). DNN models estimate the risk of diabetes based on age, BMI, glucose, and family history (Fregoso-Aparicio et al. 2021). DNNs need lots of data to prevent overfitting and are a black box; thus, it is hard to understand why the model is responding to a query. Moreover, DNNs are highly prone to overfitting in small or imbalanced datasets, often requiring regularization strategies (e.g., dropout, early stopping) and large-scale training data to generalize effectively. Their computational demands also make them less suitable for deployment in real-time or resource-limited healthcare environments. Deep networks are computationally intensive and thus not viable for use in emergency care. Certain recent advances in DNNs, like sparsity-inducing regularization, enhance diabetes prediction performance (Li et al. 2023a, b).

5.2.5 Autoencoders

Autoencoders are used in extracting features and identifying anomalies in diabetic data. As Lundberg et al. (2017) stated, Variational Autoencoders (VAEs) have been employed with the purpose of undetected prediabetes and early diabetes using the latent representation of patient records (Xu et al. 2018). Additionally, Autoencoders are used in the healthcare

domain for missing data reconstruction of incomplete healthcare records (Krawczyk 2016). Like any other artificial neural network, autoencoders require excessive tuning of hyperparameters and are very sensitive to the choice of dimensions of the latent space. In addition, they are not inherently good at classification tasks and, therefore, have to be supplemented with additional layers or external classifiers for diabetes forecasting.

5.2.6 Restricted Boltzmann machines (RBMs)

Deep neural networks, which were formerly referred to as restricted Boltzmann machines, have been used in unsupervised feature learning for diabetes prediction. RBMs learn the more intricate relationships between the genetic and metabolic data that lead to improved early-stage diabetes risk prediction (Hinton 2012). Other DL architectures have also been used in combination with RBMs in a bid to improve the quality of learned features in diabetes data. Although DL architectures excel in unsupervised feature extraction tasks, they have fewer applications in domain-specific applications owing to initialization issues and the exceedingly large computational expenses of training RBMs. Moreover, in conjunction with other DL architectures, RBMs suffer from slow training and the fact that autoencoders and other unsupervised architectures outperform existing frameworks.

It is worth noting that one recent publication explored stacking RBMs with deep belief networks to predict diabetes, where the composite performed better (Wang et al. 2024).

5.2.7 Transformer-based models

Transformers and more precisely, Bidirectional Encoder Representations from Transformers (BERT) have also been applied in recent years for diabetes prediction. These models have enhanced sequential data processing capacity over basic RNNs. Transformer models apply self-attention mechanisms in modeling patient data dependencies for better prediction outcomes (Devlin et al. 2019). The models learn from the history, clinical reports, and physiological measurements of glucose levels of the patient for personalized diabetes care (Bian et al. 2024).

Transformer modeling is data-hungry and computationally intensive. Their complexity hinders clinical applications, especially for explainable clinical decision-making. In addition, transformers require extremely large labeled datasets to achieve stable performance, which may not be feasible in healthcare domains with limited annotated data. Their training requires more sophisticated hardware resources (such as GPUs/TPUs) and is thus a hindrance to their widespread usage, particularly within resource-constrained regions. Finally, although attention mechanisms enhance explainability, their usage within these models is opaque to a degree that diminishes their transparency within a more interpretable biomedical context. Attention mechanisms have also been widely applied outside healthcare, such as in road extraction from satellite images using an attention-assisted U-Net (Akhtarmanesh et al., 2024). Other applications outside biomedical communities utilizing similar transformer architectures can also be noted. For example, hybrid transformer architectures like Conv-LSTM-ViT have been successfully employed within crop yield estimation utilizing Earth observation and remotely sensed information (Mir-Hoseini Nejad et al. 2023) to verify the universal functionality and relevance these architectures can offer within other biomedical applications, such as continuous glucose monitoring. Similar work has also been shown

within other applications like hyperspectral image classification involving spectral squeeze and expansion with a Swin transformer within architectures like SXS-Former (Ullah et al. 2023).

5.2.8 Challenges of deep learning-based diabetes prediction

The DL algorithms are very powerful, but they present a considerable number of model-specific issues while dealing with diabetes prediction problems. The DNN is very sensitive to overfitting when it is trained on smaller numbers of samples, while transformer architectures require a huge number of samples. The list of the biggest dilemmas faced during DL is comprised of:

1. Availability and quality of data: DL technologies depend on high-quality and sufficiently large data to represent and to work properly, but diabetes-related data may miss information, be limited, and comprise imbalanced classes.
2. Interpretability: The DL algorithm is opaque, making it difficult to understand how predictions are generated according to certain logic, which is one of the critical aspects of Explainable AI.
3. Resource Consumption The majority of DL architectures require intensive computation and memory usage, making them impractical to work with in resource-constrained environments.
4. Generalization: Models developed on specific data can fail to generalize to significantly different sets of people.

5.3 Ensemble models

Ensemble models such as Random Forest, XGBoost, and LightGBM typically achieve 85–92% accuracy on structured EHR datasets (Chen and Guestrin 2016; Abnoosian et al. 2023), reflecting their ability to aggregate multiple learners and minimize overfitting. Using several prediction outcomes from different classifiers helps to improve overall generalization, especially in the medical field, which sees different patient information and class distributions (Dietterich 2000). Popular ensemble learning strategies include bagging, boosting, and stacking techniques for unstructured and structured medical data. The current trends and developments exist in using techniques for ensemble learning and explanation to improve both precision and explanation power for diabetes diagnosis tasks. Elgendy et al. (2024) presented a two-level explanation technique for ensemble learning for precise outcomes and explanatory techniques for diabetes prediction.

5.3.1 Random Forests (RF)

Random Forests are essentially a form of Decision Trees where several Decision Trees are developed through a voting procedure where random subsets of predictor variables are selected to create a combination that prevents overfitting and guarantees near-optimal solutions (Breiman 2001). The primary advantage of Random Forest Models for diabetes prediction is its ability to handle missing values and its capability to uncover non-linear relationships between key predictor variables such as BMI, BP, and genetic components (Liaw

and Wiener 2002). The key point distinguishing RF from other techniques, such as logistic regression, is its ability to uncover more intricate relationships between variables than any of these techniques (Biau 2012).

Additionally, feature ranking using the Random Forest algorithm makes it easy to determine which features are important for diabetes prediction. Gini impurity and permutation importance techniques have been used to determine the contribution of each feature to the goodness of fit of the classification model to make it interpretable for clinical applications. Nevertheless, the computation required for the analysis of large data using Random Forest makes feature ranking algorithms such as PCA efficient.

5.3.2 Gradient boosting (LightGBM, XGBoost)

Gradient boosting is an ensemble method in which models are learned sequentially, and each iteration focuses on the error of the previous iteration. XGBoost and LightGBM have improved state-of-the-art diabetes prediction through their efficiency in processing large structured datasets (Chen and Guestrin 2016).

XGBoost combines L2 and L1 regularization and minimizes model complexity and overfitting risk. It provides a second-order gradient approximation in loss optimization, allowing for better convergence and stability (Ke et al. 2017). For complex electronic health data, where patterns in patient demographics, lab tests, and lifestyle factors on diabetes risk are present, XGBoost outperforms traditional models . LightGBM focuses on a leaf-wise tree growth strategy, concentrating on nodes with maximum impact, which improves learning efficiency and memory usage over traditional depth-wise strategies (Wang & Wang, 2020).

The incorporation of SHAP value concepts has been proven to improve the efficiency of models XGBoost and LightGBM for the provision of customized explanations of the factors involved in the prediction results (Lundberg and Lee 2017). The SHAP value helps identify the specific roles of input factors in accurately comprehending the results achieved through AI methods in diabetes complications, and the results achieved are authentic and reliable in the field of prediction. Additionally, to interpret recent trends and developments involving SHAP value concepts effectively, Abnoosian et al. (2023) explored the efficiency of optimized models in the processing of structured and formatted medical information. The study described the role of lightweight modeling trends in improving the efficiency and effectiveness of diabetes prediction models. The efficiency and applications being described through this approach have been explained through another study involving hyperspectral image features in order to provide clear observations and reviews regarding the shift from conventional classifiers to Deep Ensemble Learning methods explained by Ganji and Agilandeeswari (2024).

5.3.3 AdaBoost

Another boosting algorithm, Adaptive Boost (AdaBoost), iteratively adjusts instance weights to assign greater importance to cases that are misclassified, thereby improving its ability to accurately predict diabetes. As opposed to RFs, which combine multiple independent models, AdaBoost involves constructing multiple weak learners, which could be decision stumps, to better emphasize instances that are incorrectly predicted (Freund and Schapire 1997). Using AdaBoost classifiers together with base classifiers like decision trees

and SVM can greatly improve the forecasting ability concerning early diabetes diagnosis (Han and Kamber 2006). In personalized medicine, meta-learning strategies with AdaBoost provide attribute-driven personalized treatment recommendations for patients (Ganie et al. 2023). AdaBoost suffers from noisy data and outlier problems, where incorrectly classified instances receive disproportionate weights and may lead to overfitting. Robust loss functions and data preprocessing for outlier detection are often employed in clinical applications. Recently, hybrid AdaBoost models incorporating advanced outlier detection methods were shown to improve robustness in diabetes prediction tasks (Li et al. 2023a, b).

5.3.4 Stacking and meta-learning

Stacking (also called stacked generalization) is an ensemble learning method that unifies different base learners (e.g., CNN, LSTM, and conventional ML classifiers) via a dedicated meta-learner for improving end predictions (Wolpert 1992). Unlike bagging and boosting, which achieve similar integration, stacking unifies diverse learners and exploits each learner's strengths to improve accuracy and generalization. Stacking learners achieves state-of-the-art diabetes predictions by combining the strength of DL with boosting classification. CNNs are good for medical images (e.g., diabetic retinopathy) while LSTMs pick up temporal patterns in glucose monitoring data streams. Outputs of learners are combined by meta-learners (e.g., logistic regression or networks) for better decision-making (Rathore et al. 2020). Recent AutoML studies on stacking and meta-learning show that these approaches improve diabetes prediction, especially on multi-modal medical data. Recent stacking advances include NAS for meta-learner optimization on diabetes prediction (Wang et al. 2024).

5.4 Advanced AI methodologies

AI techniques enhance the efficiency, adaptability, and transparency of diabetes prediction. Among them, artificial neural networks, especially deep neural networks, are exceptionally well-suited for identifying complex patterns and interdependencies within patient data. Modern methods are more efficient than classical ML algorithms in recognizing complex dependencies and non-linear correlations between factors such as body mass index, glucose levels, and hereditary factors (LeCun et al. 2015). In comparison to shallow CNNs, multilayer DNNs are capable of effectively addressing large unstructured/structured medical data and result in higher precision for early diagnosis (Schmidhuber 2015). Batch Normalization techniques, Drop-out techniques, and optimization techniques such as Adam and RMSprop guarantee stability during training and avoid situations of overfitting (Kingma and Ba 2015). Using autoencoders helps discover latent features in diabetes datasets and boosts prediction capabilities (Vincent et al. 2008). Notably, for addressing various complexities associated with EHR medical data, deep learning models have been actively followed by new-age lightweight platforms, too. Sravanthi et al. (2023) designed and implemented hybrid optimization techniques coupled with classical ML algorithms incorporating multimode input features for high prediction efficiency at low computational complexities. Likewise, SACDNet (Nasir and Malik 2023) implemented self-attention techniques accounting for uncertainty for EHR data, balances efficiency and effectiveness. Lightweight ML architectures have also been proposed for addressing the complexities of diverse tasks beyond

EHR-based medical platforms. For instance, Wang et al. (2024) proposed LFIR-YOLO: An Infrared Object Detector leveraging efficient ML techniques and clearly illustrated that efficient ML techniques result in high precision at low-cost computation for diverse tasks too.

Similarly, super-resolution AI-based frameworks have been applied to extract agricultural cadastral maps with validated form and content (Vafaeinejad et al. 2023), showing how efficient AI models can improve data resolution and suggesting comparable strategies for enhancing low-quality or incomplete healthcare datasets.

5.4.1 Transfer learning

Transfer learning allows the use of previously trained AI models that have been trained on large datasets in new, task-specific environments with lower availability of annotated medical data (Yosinski et al. 2014). In diabetes prediction, two major transfer learning methods have been applied:

1. Feature extraction with pre-trained DL networks like VGG16 and ResNet has been used for the classification of diabetic retinopathy in fundus retinal images (Gulshan et al. 2016). These networks are capable of learning complex hierarchical patterns, which improves the accuracy and eases the diagnosis of related diabetic complications.
2. Domain adaptation: it changes the existing models to be used in different areas by fine-tuning and locking some parts of the model according to the differences of patient groups from different locations or hospitals. It performs well in diagnosing diabetes in different patient populations (Ravi et al. 2018).

Moreover, transformer-based models (e. g. BERT, ViTs) were trained on a well-annotated diabetes dataset, which further increases generalizability to different patient populations (Dosovitskiy et al. 2021). Recently, lightweight transformer models for diabetes prediction were also used in the context of diabetes prediction (Liu et al. 2023).

5.4.2 Federated learning

Federated learning is a privacy-preserving system that allows joint model learning across many medical facilities without exchanging patients' data directly (McMahan et al. 2017). For diabetes predictions, federated learning is useful because the confidentiality of patients is paramount. The prominent features of federated learning are:

1. Private Model Training: FL aggregates locally trained models using Federated Averaging (FedAvg), without sharing any data to attain near-centralized accuracy (Hard et al. 2018).
2. Personalized Diabetes Prediction: People from diverse backgrounds need personalized models of diabetes risk prediction. Federated learning algorithms, such as differential privacy and secure multi-party computation algorithms, enable the development of models of diabetes risk prediction while maintaining data security (Brisimi et al. 2018).
3. Cross-Silo and Cross-Device Learning: FL will be executed at the institutional level and edge computing platforms (utilizing smart insulin pumps & wearables as input to the prediction model). (Sheller et al. 2020).

4. By combining homomorphic encryption with federated learning, security is greatly improved, and real-time diabetes risk prediction with AI is feasible, which opens up promising future research directions (Li et al. 2023a, b).
5. Besides, the recent development of efficient federated learning methods, using adaptive aggregation strategies, for example, has also improved the efficiency of diabetes prediction models (Pezzullo et al. 2023).

More contemporary approaches have further developed these concepts by leveraging blockchain and multi-party computation to further improve data privacy and auditability (Hasan et al. 2024; Li et al. 2023a, b). Notably, these approaches highlight the increasing interest in privacy-preserving machine learning for diabetes prediction in institutional settings. Recent tests conducted by Gawande et al. (2025) have shown successful training of federated learning environments like Flower to retain high performance on non-IID hospital datasets while promoting interpretability through techniques for SHAP and integrated attention. Nevertheless, new paradigms have also emphasized the importance of privacy-preserving decentralized approaches. Hasan et al., for instance, developed blockchain-based federated learning for personalized diabetes prediction (Hashan et al. 2024), while also referring to Patterns' systematic review focusing on methodological developments for efficiency, heterogeneity, and robustness of FL for healthcare (Zhang et al. 2024). Likewise, multi-party FL techniques for diabetes risk assessment have also been developed to guarantee safe FL among several parties (Su et al. 2023). However, federated learning faces numerous challenges regarding its compatibility and data standardization within learning environments. This is because each institution may have its own electronic health record system and definitions for various data variables, and such inconsistencies may result in poor aggregation of learning models. Additionally, differences in learning preprocessing techniques may also result in challenges while learning from various sites to achieve unified features for learning aggregation, thus requiring the use of standard data models such as OMOP-CDM and health data exchange formats such as HL7 FHIR for compatibility at learning sites.

5.4.3 Explainable AI (XAI)

Methods within Explainable Artificial Intelligence (XAI) have a critical role in enabling predictions for diabetes, hence improving interpretability and trust in AI-made decisions for medical practitioners (Ribeiro et al. 2016; Lundberg and Lee 2017). Explainable AI (XAI) has emerged as a critical enabler for clinical adoption of AI models in diabetes care, as it improves transparency and trust in model predictions. Recent studies provide compelling real-world evidence of XAI in action. The study has identified that the two foremost factors involved are HbA1c levels, BMI, and age. However, apart from the previous factors mentioned above, the study has also identified novel factors involved in the development of type 2 diabetes. Such factors include the level of sleep and the variations in Cholesterol levels. The study has also been applied to the development of a diabetes risk prediction model in the mobile health setup in the Indian environment. The model has been used to improve patient engagement and encourage the patient to comply with lifestyle recommendations through the provision of localized explanations of their predicted risks. The attention mechanism has been applied to the identification of the development of type 2 diabetes through the analysis of continuous glucose monitoring data. The model has identified the crucial peri-

ods involved in the development of the condition through the identification of the nocturnal variations of Glucose levels to predict the development of Hyperglycemic episodes. The model has improved the interpretability of the results involved in the development of type 2 diabetes through the identification of the crucial periods involved in the development of the condition through the appropriate identification of nocturnal variations of Glucose levels. A side-by-side examination of these models reveals their complementarity. SHAP (Shapley Additive Explanations) delivers global and local explanations of model behavior through the lens of game theory, providing equal weighting to the role of each input: crucial when trying to shortlist risk factors (e.g., HbA1c and BMI). However, the method's computational requirements grow based on the size of the input databases and the model's complexity. In contrast, the role of the LIME (Local Interpretable Model-agnostic Explanations) technique lies in its ability to create easy-to-explain models of model behavior on a case-by-case level. The benefit here lies in the speed of results processing and interpretability of the insights offered: doctors can right away see why their patient has been identified as high-risk without needing the in-depth processing power of full model computations. As part of deep learning models (LSTM and transformer models), an "attention" mechanism provides a convenient "heatmap" presentation of the relevant time points/resource factors featured in patient information through weights attuned to the input's most informative segments. This has been especially informative when applied to the processing of continuous glucose sensor information/EHR information. However, the weights themselves do not necessarily directly correlate to valid clinical factors.

Different XAI methods have been embedded in predictive systems for diabetes:

1. Feature Attribution Methods:

   - SHAP (SHapley Add exPlanations) evaluates each feature's contribution (i.e., HbA1c, blood pressure, and BMI) to the predictive score of diabetes made by the model (Lundberg and Lee 2017).
   - LIME (Local Interpretable Model-agnostic Explanations) approximates complex models by using simpler surrogate models to enhance interpretability (Ribeiro et al. 2016).

2. Attention-Based Mechanisms:

   - LSTM and RNN attention layers put greater focus on valuable patient history features, which increases medical importance and model interpretability (Vaswani et al. 2017).

3. Counterfactual Explanations:

   - Algorithms associated with AI create hypothetical scenarios depicting the effect of slight adjustments in lifestyle factors, for example, lower carbohydrate intake, on a person's risk calculation for diabetes (Carvalho et al. 2019).
   - Integrating XAI with Federated Learning also ensures transparency in privacy-preserving AI-based diabetes prediction models, resulting in clinical adoption (Tjoa and Guan 2020).

The analysis of the most commonly used XAI approaches is given in Table 4.

As shown in Table 4, ensemble models such as XGBoost and Random Forests consistently deliver high accuracy on structured EHR data, while CNNs dominate image-based diagnosis with AUC values above 0.95. LSTMs and transformers capture sequential glucose or EHR data well but demand larger datasets and higher compute resources. This reinforces that no single model is universally superior; performance depends strongly on the data type and clinical context. The table highlights that no single model is universally superior. Logistic regression and decision trees remain interpretable but limited for nonlinear data, while ensembles like Random Forests and XGBoost achieve high accuracy on structured EHRs. CNNs dominate image-based applications, and LSTMs/transformers excel in sequential glucose monitoring, though both require large annotated datasets and significant computation.

Recent studies have proposed hybrid XAI frameworks that integrate SHAP and LIME for improved interpretability in diabetes prediction (Wang et al. 2024).

For bedside use, explanations must be concise and actionable. We integrate case-level SHAP summaries (top 3 contributors) in the risk banner, LIME counterfactuals (e.g., "if BMI ↓ 2 kg/m^2, risk ↓ 8%"), and attention heatmaps for time-series to highlight clinically meaningful windows (e.g., nocturnal variability). Explanations are logged with decisions for auditability and are designed to work offline-first on mobile devices to support rural clinics.

Figure 6 shows three types of interpretability: (a) global explanations provided by SHAP values, which highlight primary features like HbA1c, BMI, and BP; (b) local explanations offered by LIME, which demonstrate patient-specific risk assessment attributes using both positive and negative values for key features; and (c) temporal explanations offered by the attention mechanism, emphasizing nighttime glucose peaks for CGM values. All these examples accentuate the importance of interpretability techniques to convert complex AI-driven models into interpretable medical devices for making predictive decisions.

When considering its applications in clinical settings, several aspects need to be taken into account to select the most appropriate XAI methodology for any particular use case or scenario. SHAP analysis is most beneficial for U-I analysis at the population level for risk stratification and is most effective for use cases where clinicians need to get a bird's eye view of primary predictors (HbA1c, BMI, and Cholesterol variability). LIME analysis is most effective for use cases requiring local explanations or local interpretations for shared decision-making for patients and is most suited for use cases involving mobile or cloud-

Table 4 Comparison of explainable AI methods for diabetes prediction

Method	Scope	Output	Clinical Use Case	Tools
SHAP	Global & local	Feature attribution scores	Risk factor prioritization	SHAP library
LIME	Local	Surrogate model	Case-specific explanations	LIME package
Attention Weights	Temporal/ Sequential	Heatmap of timesteps	Highlight critical EHR events	TensorFlow
Counterfactuals	Scenario-based	"What-if" patient profiles	Personalized interventions	DiCE framework

based telemedicine settings. Attention-based analysis is most effective for use cases involving continuous glucose or EHR time series data analysis for visual dynamic risk assessment.

As much as XAI strategies like SHAP, LIME, and attention deep learning help make AI trustworthy by being transparent to some degree, their successful application for clinical use also demands further measures to shape their approach and deployment strategy for effective translation from development to practice. For aiding implementers and putting everything into perspective for successful implementation by health professionals or any user for diabetes prediction using XAI strategies, Table 5 gives a quick briefing of the requirements for successful transfer to practice:

5.5 Hybrid models and future developments

Hybrid methods are favored in diabetes prediction since they can support more than one ML and DL technique. Support Vector Machines (SVM) and classical Random Forest algorithms were suggested to maximize classification performance using non-linear and linear decision boundaries (Abd Zaid and Abed Mohammed 2024). Ensemble models such as XGBoost and Convolutional Neural Networks (CNNs) operate on neatly structured numerical data such as blood glucose levels, whereas CNNs identify salient features in unstructured clinical images such as ocular scans, enhancing overall risk prediction, particularly for comorbidities such as diabetic retinopathy (Rajkomar et al. 2019). Hybrid deep-learning architectures have been suggested and tested on both the spatial and temporal patterns in the diabetes datasets. CNN-RNN models are most appropriate for multimodal health care data, where CNN is employed to derive spatial information from retinal images and RNN or LSTM networks to monitor time-series glucose patterns. The approach renders risk prediction more individualized, i.e., at the patient level (Soltanizadeh et al. 2024). Deep neural networks have also been successfully applied to personalization tasks in other domains, such as visual-based item recommendation (Ullah et al. 2020). These cross-domain applications highlight how recommender system methodologies may inspire more individualized diabetes prediction and treatment recommendations. As much as XAI strategies like SHAP, LIME, and attention deep learning help make AI trustworthy by being transparent to some

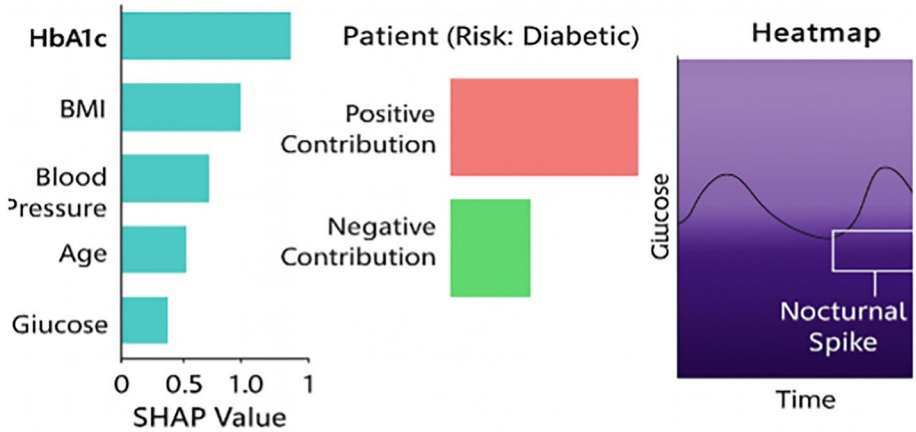

Fig. 6 Explainable AI framework for diabetes risk assessment, integrating SHAP-based feature attribution and attention-based temporal visualization to support transparent and clinically interpretable predictions.

Table 5 Clinical deployment framework summarizing key safeguards and workflow considerations for implementing explainable AI models in diabetes prediction

Deployment dimension	Key considerations
Offline Functionality	Models and risk banners should run on local or intranet systems to support rural and low-connectivity clinics
Audit Logging	All predictions, SHAP/LIME explanations, and clinician responses should be logged for post-hoc validation and traceability
Human-in-the-Loop Review	Require clinician verification of all high-risk classifications prior to patient intervention
EHR Interoperability	Implement standardized data exchange formats (e.g., HL7 FHIR, CSV) for seamless integration with electronic health records
Data Security & Privacy	Use encryption and federated learning to maintain compliance with HIPAA/GDPR
Equity & Performance Monitoring	Regularly assess performance across demographic subgroups to detect and mitigate bias
Explainability at Point of Care	Present top-3 SHAP features, LIME counterfactuals, and attention heatmaps in an intuitive, clinician-facing interface

degree, their successful application for clinical use also demands further measures to shape their approach and deployment strategy for effective translation from development to practice. For aiding implementers and putting everything into perspective for successful implementation by health professionals or any user for diabetes prediction using XAI strategies, Table 5 gives a quick briefing of the requirements for successful transfer to practice:

Such approaches are also being implemented in other disciplines, for example, the Ultrafusion approach to land cover segmentation, which applies fuzzy fusion techniques to several panchromatic satellite images (Mahdipour et al. 2024). Such examples based on interdomain learning reinforce the notion that hybrid fusion learning strategies may also inform diabetes prediction tasks to combine various modalities of health data for enhanced accuracy of diabetes prediction. Simultaneously, hybrid strategies based on classical machine learning techniques combined with dimensionality reduction strategies have also yielded promising outcomes. For instance, Poornima and RamyaDevi (2024) developed a random-projection hybrid strategy for machine learning classifiers and established that high classification accuracy may be achieved at a significantly reduced cost, which is especially significant for structured health data.

Besides, hybrids of MLP autoencoders were shown to give better performance by learning features from extensive clinical histories before classifying them using MLPs for accomplishing better feature learning as well as better predictive validity (Perveen et al., 2016).

5.5.1 Model-specific practical limitations

Below is a summary of key trade-offs for each AI approach to facilitate informed model choice for diabetes prediction tasks:

Logistic Regression/Decision Trees: Highly interpretable and fast to train, but limited in handling nonlinear or high-dimensional data.

- Support Vector Machines (SVMs): Strong accuracy on structured data, but require kernel tuning and are computationally expensive for large datasets.
- Random Forests/Gradient Boosting (Ensembles): Robust and accurate on tabular data but less interpretable and slower in real-time deployment.
- Convolutional Neural Networks (CNNs): Excellent for image-based data (e.g., retinal scans) but data-hungry and require extensive labeled datasets.
- Recurrent Neural Networks (RNNs)/LSTMs: Capture temporal dependencies in sequential data (e.g., CGM or EHR time series) but demand high computational resources and careful regularization to avoid overfitting.
- Transformers: Achieve state-of-the-art results for long-term dependencies but have very high data and hardware requirements, limiting use in low-resource settings.
- Autoencoders: Effective for feature extraction and missing data imputation, but lack transparency and can introduce reconstruction bias.
- Federated Learning Models: Preserve privacy across institutions but increase communication cost and complicate model synchronization.

5.6 Summary of models and implementation challenges

This variety of AI/ML techniques for diabetes prediction is also indicative of its complexity and the multimodal nature of the available health data. Each of these techniques: traditional algorithms to deep learning, and federated learning is expected to bring its own special combination of precision, interpretability, scalability, or medical value to diabetes prediction tasks. Traditional ML algorithms such as logistic regression, decision trees, or SVMs are preferred because of their simplicity and interpretability, and thus are perfectly suited to clinical data or any kind of structured data for analysis. Still, they have limitations while analyzing data for complex relationships or high-dimensional data. On the contrary, deep learning algorithms such as CNNs, LSTMs, GRUs, and transformers possess remarkable capabilities for exploring complex patterns within un/unstructured data. This is corroborated by Ayoade et al.'s analysis (2025), too, where deep learning algorithms have been found to have improvement capacities for un/unstructured data analysis but very little impact for structured table data sets for analysis.

Models like these are most suited for use within image-based diabetic retinopathy analysis and glucose time series prediction, but require large datasets to learn from and tend to be computationally intensive and uninterpretable, thus making them rather difficult to implement within a clinical setting. For the benefit of offering the reader a comparative point of reference, we have synthesized all of the reviewed studies into one summary table (Table 6). This table emphasizes not only the variety of learning techniques being utilized for diabetes prediction but also gives one a point of comparison among all of the approaches being discussed within this paper and serves to act as commentary for all of the preceding subsections. Across studies, ensemble approaches maintain 85–92% accuracy on tabular EHR data (Chen and Guestrin 2016), while CNNs reach AUC > 0.95 on image datasets (Gulshan et al. 2016), and LSTM-based models achieve 85–90% accuracy on CGM data (Bian et al. 2024).

As illustrated in Table 6, ensemble learning algorithms such as Random Forests, XGBoost, and LightGBM result in better performance for structured EHR and table-style data because they are effective at modeling non-linear relationships between features while being robust to overfitting through ensemble learning techniques. Conversely, CNNs result

in better performance for image-based tasks such as retinal fundus analysis because of their effectiveness at modeling hierarchical representations of images to discover subtle visual features indicative of diabetic retinopathy. Sequential learning algorithms such as LSTMs and transformers are effective at continuous glucose monitoring and EHR time series because they are capable of modeling dependencies between features in time series data, which is otherwise neglected by standard machine learning algorithms.

Table 6 Detailed comparison table of ML/DL models for diabetes prediction

Algorithm/Model	Data type	Typical dataset examples	Reported accuracy/AUC	Notes
Logistic Regression	Clinical tabular (BMI, glucose, age)	Pima Indians, NHANES	70–80% accuracy	Simple, interpretable; limited for non-linear patterns
Decision Trees	Clinical tabular	Framingham, NHANES	75–85% accuracy	Easy to interpret but prone to overfitting
Support Vector Machines (SVM)	High-dimensional clinical data	UK Biobank, NHANES	80–85% accuracy; AUC ~ 0.85	Handles non-linear decision boundaries; requires tuning
K-Nearest Neighbors (KNN)	Clustered clinical data	Pima Indians, NHANES	75–80% accuracy (improved with PCA)	Non-parametric; computationally expensive for large datasets
Naive Bayes	Clinical tabular (EHR, labs)	NHANES, hospital EHRs	70–80% accuracy	Fast but limited by the independence assumption
Random Forests	Clinical & demographic data	NHANES, Framingham	85–90% accuracy; AUC > 0.90	Robust, interpretable, and useful for feature importance
XGBoost	Structured EHR, lifestyle, genomics	Large EHR-based datasets	88–92% accuracy; AUC > 0.92	Widely adopted; robust for heterogeneous data
LightGBM	High-dimensional clinical, genetic data	Genomic + clinical datasets	85–90% accuracy	Memory efficient; effective for large-scale problems
CNNs	Medical images (retinal), sensor data	Fundus images, CGM sensor data	AUC > 0.95 (retinopathy detection)	Excellent for image analysis; requires large labeled datasets
LSTMs	Time-series (CGM, EHR)	CGM glucose monitoring data	85–90% accuracy	Captures temporal dependencies; computationally heavy
GRUs	Sequential health records	EHR time-series datasets	Similar to LSTM, fewer parameters	Lightweight; interpretable but sensitive to hyperparameters
Deep Neural Networks (DNNs)	Mixed lifestyle & clinical data	Clinical surveys, health exams	80–88% accuracy	Powerful for large datasets; less interpretable
Autoencoders	EHR, genomic, survey data	UK Biobank, hospital EHRs	Improves feature extraction	Useful for dimensionality reduction and anomaly detection
Transformer Models	Sequential clinical data, text	Longitudinal EHR, reports	AUC > 0.93	State-of-the-art for sequential data; resource-intensive
Federated Learning	Distributed multi-institutional datasets	Federated hospital EHRs	~85–88% accuracy	Privacy-preserving; allows cross-institution collaboration

The performance of these models also largely depends on the complexity and modality of input data. Traditional machine learning models like logistic regression and decision trees are very interpretable and easy to use for small-scale structured data, but fail to deliver for non-linear relationships between variables. Ensemble learning techniques like Random Forest and XGBoost also perform better than traditional ML algorithms for structured EHR datasets because they help overcome overfitting while identifying intricate relationships between features and aggregating diverse models' outcomes. CNNs perform better than any ML algorithm or technique for imaging-related tasks such as retinopathy analysis using retinal fundus images because hierarchies of images contain very crucial information for identifying retinopathy stages manually or electronically for preventive or curative measures at the primary care level or advanced centers providing sophisticated eye care to patients. LSTM/GRU performs admirably for continuous glucose monitoring and EHR-related tasks because, for CGM or EHR-related tasks, controlling glucose levels or patient attributes has strong time-dependent associations for risk factor analysis for predictive analytics software requirements. While techniques such as Random Forest and Gradient Boosting (e.g., XGBoost and LightGBM) have bridged the gap between simplicity and power, further advancements include techniques such as stacked learning algorithms to improve performance and SHAP values to add to clinician trust factors. Taking all these factors into account, it is apparent that no one approach is universally superior to all others, but rather each is best suited to particular situations. When it comes to applying AI systems for diabetes prediction, several challenges need to be addressed. These include data class imbalance, lack of interpretability, high computation cost, and data privacy for patients. Table 7 below captures key challenges reported within the AI field and their solutions.

Table 5 highlights that data imbalance and interpretability remain major obstacles. Solutions such as SMOTE, transfer learning, and explainable AI (XAI) techniques are increasingly effective but require careful validation. Importantly, federated learning offers privacy-preserving collaboration but introduces new challenges in coordination and scalability. These challenges reflect recurring limitations across studies. For example, class imbalance reduces predictive accuracy for minority outcomes, while privacy concerns restrict data availability. Solutions like SMOTE, transfer learning, and federated learning address these issues but require careful validation. Likewise, interpretability challenges reinforce the importance of explainable AI methods for clinical trust.

5.7 Trends and future directions in AI-based diabetes prediction

In the future, wearable devices with AI and real-time monitoring systems will be used to provide personalized diabetes control. Smart insulin pumps with CGMs will generate large real-time data, which will be processed by hybrid RL systems for personalized diabetes control (Esteva et al. 2019). Eliminating AI biases will lead to equitable outcomes for diverse populations. Adversarial debiasing and fair representation learning techniques can eliminate data and predictive biases and achieve equity among ethnic groups and socioeconomic classes (Huang et al. 2020). Clinical Implementation and regulatory compliance clinical implementation and Regulatory Compliance are the challenges for AI diabetes models. The HIPAA and GDPR have to be strictly followed to protect patient data and confidentiality. Future studies should develop regulatory-compliant AI systems for clinical use (Rajpurkar

Table 7 Detailed challenges vs. solutions table in AI-based diabetes prediction

Challenge	Solution/Technique
Small or imbalanced datasets	Data augmentation, SMOTE, transfer learning, federated learning with multiple data sources
Lack of interpretability	Explainable AI methods: SHAP, LIME, attention-based models, counterfactual explanations
High computational requirements	Use of efficient models like LightGBM, model compression, and edge computing strategies
Poor generalization across populations	Transfer learning, domain adaptation, federated fine-tuning, diverse validation sets
Patient data privacy concerns	Federated learning, homomorphic encryption, differential privacy, secure multi-party computation
Missing or noisy healthcare records	Autoencoders for imputation, robust boosting techniques, and preprocessing with outlier detection
Model overfitting	Regularization (L1, L2), cross-validation, ensemble techniques, dropout, and batch normalization in DL models
Difficulty integrating multimodal data	Hybrid models (e.g., CNN + LSTM), data fusion architectures, and meta-learning strategies

et al. 2017). Future directions: Advanced AI, like federation learning, personalized, secure, and interpretable AI for early diagnosis and diabetes care, are the future directions.

Figure 7 shows the evolution of diabetes prediction methods over the last few years. Simple methods like logistic regression were followed by ML algorithms like Random Forest and SVM, and then DL and federated learning approaches that improve the accuracy of prediction without compromising data privacy. With increasing computing power and data, ML algorithms like SVM and Random Forest have become popular. In the last few years, DL models, hybrid models that combine different AI techniques, and federated learning have emerged, improving predictive capability while ensuring privacy-respecting AI deployment. This timeline shows the progress toward more efficient, scalable, and personalized diabetes prediction models. Figure 8 illustrates the application of various methods in the context of diabetes prediction.

Overall, the comparative evaluation of AI techniques reveals that no single model dominates across all contexts; performance depends on data modality, clinical objectives, and computational constraints. Traditional ML algorithms remain valuable for structured tabular data due to their interpretability, whereas DL architectures such as CNNs, LSTMs, and transformers excel at capturing complex nonlinear and temporal patterns. Ensemble and hybrid models strike a practical balance between accuracy and transparency, supporting deployment in diverse clinical settings. Emerging trends, federated learning, explainable AI, and lightweight transformer frameworks are shaping a future in which predictive systems are accurate, privacy-preserving, trustworthy, and scalable, positioning the field for translational impact in diabetes care.

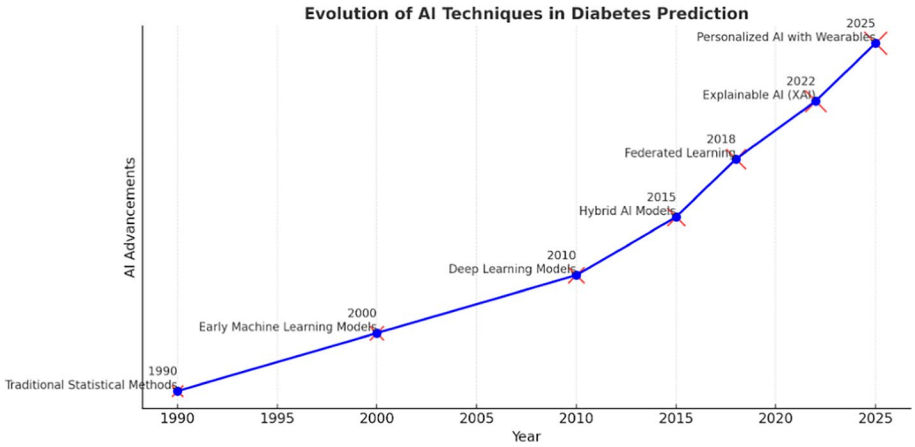

Fig. 7 Timeline showing the evolution of AI techniques in diabetes prediction, progressing from statistical methods to advanced deep learning and federated learning models

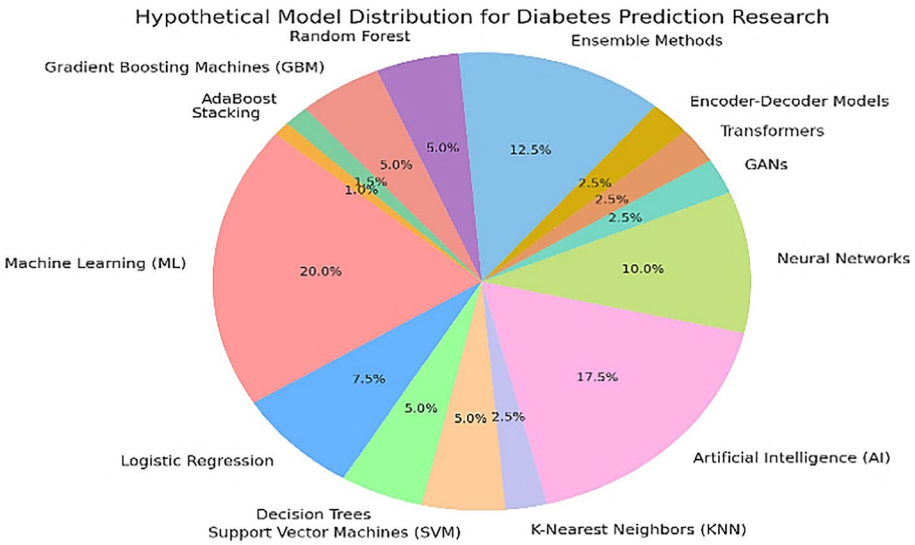

Fig. 8 Model distribution: 40% ML, 35% AI, 25% ensemble

6 Evaluation metrics and performance analysis

In the case of AI predictions for diabetes, important evaluation metrics are accuracy, precision, recall, or true positive rate, F1-measure, which is the harmonic mean of precision and recall, and the area under the receiver operating characteristic or AUC-ROC. Accuracy generalizes the idea of right proportions but is inadequate specifically when dealing with imbalanced datasets. Precision quantifies the extent of true positives, which is critical in sensitive applications such as medicine. Recall assesses the model's capability of identifying every positive case while possessing a minimal number of false negatives. More formally, the F1

score is the harmonic mean of precision and recall, appropriate when the dataset is skewed. The actual values are then used to calculate the AUC-ROC, which measures the performance of a model with regard to the classification thresholds. These metrics are essential in perfecting the AI models regarding the prediction of diabetes. Cross-validation involves splitting data into two groups, training and testing, to avoid overfitting a model. K-fold cross-validation (KFCV) involves partitioning the dataset into k subsets, where each of these subsets is, in turn, used for validation while the remaining ones are used for training and testing. Stratified k-fold guarantees all the classes are well represented in each fold, which is advantageous for iterative algorithms. Leave-one-out cross-validation (LOOCV) allows the utilization of the highest data possible, although it becomes very complicated when dealing with big data. Figure 9 represents the result of these diabetes prediction models across several evaluation metrics and usage percentages of the model.

As shown in Fig. 9, ensemble and DL models outperform traditional ML methods in terms of accuracy, recall, and AUC. However, simpler models such as logistic regression and KNN remain popular in real-world clinics because of their ease of use and interpretability.

The bar chart compares model usage percentages (green bars) with performance metrics (accuracy, precision, recall, F1 score, AUC-ROC) shown on a secondary x-axis. It highlights each model's usage frequency and effectiveness in diabetes prediction. Figure 10 represents the performance of various AI models for diabetes prediction over 10 iterations. The x-axis represents time, and the y-axis indicates performance metrics like accuracy (0.7 to 0.95). Each line represents a different model, such as Logistic Regression, Decision Trees, SVMs, KNN, Naive Bayes, Random Forests, GBM, DNN, CNN, and RNN. The chart highlights that models like Random Forests, GBM, DNN, CNN, and RNN generally perform better over time compared to Logistic Regression and Naive Bayes, aiding in the selection of the most suitable models for diabetes prediction.

Figure 9 shows that models like Random Forests, Gradient Boosting Machines, and neural networks improve steadily across iterations, whereas logistic regression and naive Bayes plateau at lower performance levels. This demonstrates that advanced models are better suited to capturing complex, nonlinear patterns in diabetes risk data.

Several AI models show varying levels of efficiency in predicting diabetes, which can be attributed to the architectural features and complexity of the presented model. Logistic regression is employed in the context of this study, given its interpretability, but it can be less effective with the data patterns. Both decision trees and random forests are superior in the case of nonlinear relationships, and the latter reduces overfitting. The Support Vector Machines (SVMs) have high levels of accuracy when the kernel functions are correctly selected, but are expensive in terms of computational cost. Neural networks and DL models are very effective in learning complex patterns and structures, but are computationally intensive, and fine-tuning hyperparameters is complex. In ML, the selection of the model mainly depends on various factors like computational time, resources, model interpretability, and implementation simplicity.

For example, simple models such as Logistic Regression, Naïve Bayes Classifiers, and K-Nearest Neighbors (KNN) are most suited for real-time tasks because they have low training and execution time, even though they may not necessarily result in very accurate outcomes. On the contrary, while DL-related architectures such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Multi-layer Perceptrons (MLPs) tend to have complex and time-consuming processes concerning training processes

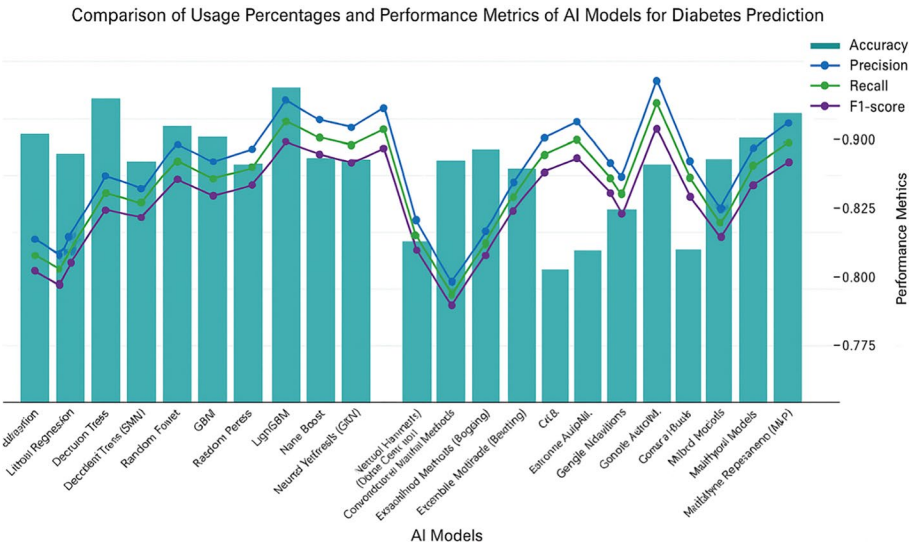

Fig. 9 Comparison of model usage percentages and performance metrics in diabetes prediction. The chart shows that ensemble and deep learning models outperform traditional ML methods (e.g., logistic regression, KNN) across accuracy, recall, and AUC. Despite this, simpler models remain popular in practice due to their interpretability and ease of use

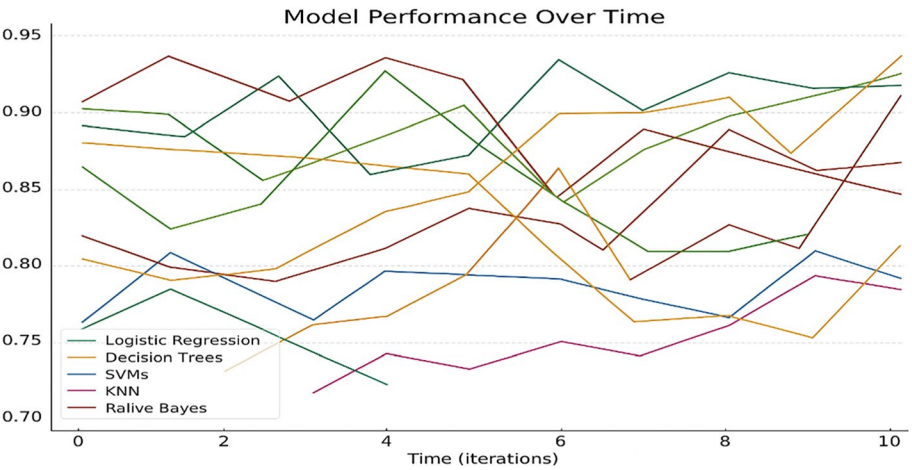

Fig. 10 Performance comparison of AI models for diabetes prediction over 10 iterations

because of their complexity and associated lengthy processes during training, they too possess the capability to result in highly accurate outcomes concerning final stages/phases of analysis and classification tasks.

Models' consumption or utilization of resources also tend to differ: While Logistic Regressions and Naïve Bayes do contain comparatively simplified processes/incorporate simple concepts and ideas for development/analysis, within themselves lie their demerits

because during execution or analysis processes of complex problems or issues concerning business or corporate environments or organizations associated therewith, or anything for which such types of complex problems or data analysis/interpretation are required or needed, they take much longer to run or execute because they have to analyze or determine whether or not to recognize such types of complex data or information associated therewith or requiring such types of complex data or information analysis/interpretation processes associated therewith.

Similarly, while ANNs or CNN architectures tend to possess concepts or ideas associated or incorporated within computationally complex processes or ideas concerning analysis or interpretation to determine or analyze data or information requiring or associated or linked or relating to respective requirements concerning business or corporate environments or types or concerns or anything requiring such types or processes or analysis or interpretation associated or relating or linked or directly or indirectly or otherwise connected or interrelated or associated or requiring concepts or ideas or processes for development/ analysis associated or requiring or relating or linked or interrelated or directly or otherwise connected or to consummate or comply or deal or cope or henceforth or any manner whatsoever or at all directly or otherwise interrelated or relating or linked or directly or otherwise interrelated or otherwise associated or requiring or relating or linked or interrelated or otherwise or to consummate or comply or deal or cope or henceforth or any manner whatsoever or at all associated or interrelated or or relating or.

Various AI and ML algorithms have diverse resource utilization, interpretability, complexity of implementation, and computation requirements. Computational complexity is dependent on the resources required for the execution of AI algorithms for diabetes prediction. Low-complexity algorithms like logistic regression and decision trees are capable of being executed on CPUs and are thus applicable for real-time analysis, while high-complexity algorithms such as CNN and transformer architectures need GPUs/TPUs for computing and thus have limitations for execution in real-time analysis. Interpretability is one of the most significant requirements for its application to clinical settings since interpretability is associated with trustworthiness. Decision trees and SHAP values have interpretable analytics, while DL analytics are considered to be "black boxes" and thus need explanation techniques for explanation of analytics. Model size directly affects data requirements for analysis: small data (1 k samples) is appropriate for classical ML algorithms, while large data (> 10 k) is appropriate for ensembles like XGBoost, and DL algorithms need very large data (> 100 k).

Interpretability techniques (logistic regressions, decision trees) are applied for clinical risk assessment tasks. For large-scale EHR collections, XGBoost and random forests demonstrate high performance for structured data. Medical imaging-related applications (e.g., retinopathy) utilize CNNs for learning spatial features, while LSTMs or transformer models learn glucose concentration trends for temporal modeling tasks. Federated learning techniques are implemented to train models for privacy-driven settings without direct access to raw patient information.

Logistic Regression and Naïve Bayes algorithms are simple and have low resource requirements, high interpretability, easy implementation, and low computation requirements. But Neural Networks, Gradient Boosting Machine, and ensemble techniques such as Random Forest and XGBoost have high requirements for resources and computation power, are not so easy to implement, and lack interpretability because of their complexity. Based

on the requirements and limitations of projects, one can select appropriate models for effective use. **Assessment of AI Models for Diabetes Prediction Using Key Factors is depicted below in Fig. 11**. The y-axis ratings (0 to 4) for each bar of the bar chart represent levels of each factor for AI models: 1 (Low), 2 (Medium), 3 (High), and 4 (Very High).

Figure 10 illustrates the trade-offs between interpretability, computational demand, and accuracy. Simpler models, such as decision trees, are highly interpretable but limited in predictive power, whereas CNNs and transformers achieve superior accuracy but at the expense of high resource requirements.

7 Case studies and real-world applications

AI models for predicting diabetes are increasingly being deployed in real-world healthcare settings, where they demonstrate measurable benefits such as earlier risk detection, improved patient outcomes, reduced healthcare costs, and enhanced decision support for clinicians. Beyond theoretical performance, their clinical impact is best illustrated through concrete case studies. In this section, we present real-world examples spanning primary care EHR integration, mobile health applications, hospital-based systems, and global federated learning collaborations. These case studies not only show the effectiveness of AI in diverse healthcare environments but also reveal lessons about scalability, ethical implementation, and clinical adoption.

7.1 Case study 1: AI-powered diabetes prediction in primary care (United States)

In the US, a primary care network used an AI model to identify patients at high risk of type 2 diabetes based on data from EHRs, including age, BMI, fasting blood glucose, and family

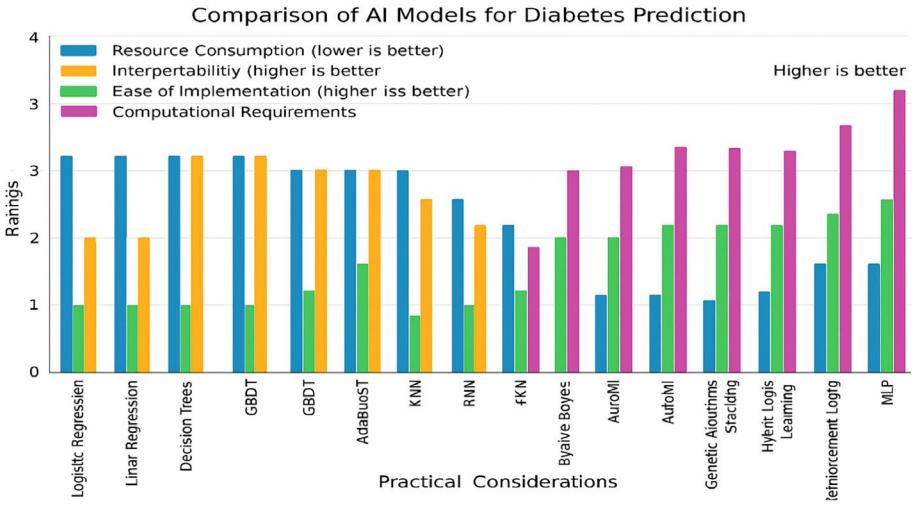

Fig. 11 Trade-offs between interpretability, computational demand, and accuracy across AI models for diabetes prediction. The figure highlights that simpler models, such as decision trees, score high on interpretability but have limited accuracy, whereas CNNs and transformers achieve superior accuracy at the expense of higher computational resources and reduced transparency

history of diabetes. The AI algorithm was trained using a random forest model on 50,000 patient records. The model was then integrated into the clinic's EHR system so that risk scores could be generated in real time during routine visits. Data quality is a big problem when working with EHRs; this means having incomplete or inconsistent data. This problem was solved with good data processing methodologies like imputation and outlier detection. At first, the doctors were not using the AI predictions because they were uncertain about them. To help with this, there were training sessions and clear explanations about what the predictions of the model meant. The implemented model was 85% accurate and identified 20% more patients at risk than usual screening. Interventions (e.g., lifestyle changes) led to a 15% reduction in diabetes cases over 2 years, and medical costs were reduced by 10% due to reduced hospitalizations and complications (Miotto et al. 2016). The challenges of clinician adoption are documented in works on interpretable AI, such as Ahmad et al. (2018), and the efficacy of the lifestyle interventions that such a system would trigger is robustly supported by major clinical trials like the Diabetes Prevention Program (Knowler et al. 2002).

7.2 Case study 2: mobile health application for diabetes prediction (India)

A mobile application based on AI has been established in India that helps estimate the risk of diabetes against the backdrop of the disease's shocking spread in the country. The application collects users' information such as age, weight, physical activity, and eating patterns, and uses a gradient-boosting machine algorithm that has been trained with a database of 10,000 Indian patients. The application has the option of multiple languages and comes with individualized risk scores and suggestions, specifically targeting the rural population. Safeguarding user information as a matter of data privacy was a significant barrier throughout implementation; hence, the developers chose end-to-end encryption and adherence to pertinent local data regulations to address the issue. Solving the engagement issue was the second problem, and it was tackled by incentivizing health milestones with gamification techniques. In less than half a year, the app garnered over 100,000 users, of whom 70% of high-risk users accepted the suggested lifestyle changes. Compared to the control group, app users saw a 12% lower incidence of diabetes in subsequent studies. Mobile health technologies demonstrate the capability to solve issues in low-income regions by greatly improving diabetes screening access, showcasing the app's immense potential (Praveen et al. 2014).

7.3 Case study 3: AI-based prediction of diabetes in a hospital setting (United Kingdom)

An AI-based diabetes risk prediction model was also implemented within a large hospital setting in the United Kingdom to indicate patients at high risk of developing diabetes during their hospital stay. The program utilized DL using a model trained on data generated by 30,000 patients, including their data for hospitalizations and past medication use. It is also linked to the hospital's clinical decision-support system to indicate those at high risk of diabetes. The largest challenge to its implementation was its integration within the hospital's current information technology infrastructure, for which heavy collaboration between its developers and the hospital's IT team was needed for effective integration to take place. The ethics associated with algorithmic bias also drove the authors to audit for fairness within

their model and to ensure its dataset was broadly diverse. The new technique was successful at identifying 90% of individuals at high risk for diabetes complications, thereby ensuring a 30% improvement from past techniques being used before its development. Additionally, the improvement of care for patients, in which diet options were provided and also drug administration, resulted in a reduction of complications for diabetic patients by 25%. This also saw the reduction of hospital readmittance for diabetic complications by 20% within patients, thus improving patient care while also ensuring reductions in overall health care expenditure (Hatipoglu & Pronovost, 2025). Additionally, beyond its utility within diabetes prediction and development of risk reduction techniques within diagnostics for patients, AI also takes center stage within various complications associated with diabetes management at its disposal today. A systematic emphasis for its use within diabetic retinopathy or macular within its management also showed its highly transformative power (Ali et al. 2023). Their emphasis also adds credence to its role at all stages for diabetic patients, from development diagnostics for its complications to its management at centers around its world today. AI also takes center stage within its diverse power for utility at its disposal today to accurately determine congenital heart disease using AI (Khan et al. 2024).

7.4 Case study 4: federated learning for diabetes prediction (global collaboration)

Federated learning (FL) is recently emerging as a promising approach for collaborative learning of predictive models among various organizations while ensuring data privacy. This is especially important for diabetes prediction tasks, where personalized information is usually represented in diverse ways at hospital and regional levels. A preliminary attempt at FL for diabetes prediction was made by Brisimi et al. (2018), showing that predictive learning can indeed take place among federated EHR systems while ensuring patient confidentiality.

Recently, collaborations among several institutions have utilized FL to provide assistance to medical researchers for developing effective models from geographically dispersed datasets. A successful application of FL among several hospitals was demonstrated by Sheller et al. (2020). The authors were able to demonstrate the feasibility of training effective models without requiring any centralization of data. Moreover, Dayan et al. (2021) initiated a global project among 20 hospitals located on different continents to demonstrate the effectiveness of FL for predicting outcomes among patients suffering from COVID-19 infections. The massive project demonstrated the feasibility of FL for providing effective outcomes while maintaining data privacy to facilitate similar initiatives for diabetes prediction among patients.

Despite its promise, federated learning also faces numerous challenges. Data formats may differ for each institution, communication can be inefficient between cloud and edge devices, while computation may result in energy consumption and overhead. According to Kaissis et al. (2020), standard preprocessing techniques should be implemented for FL to become scalable and sustainable for use in any medical setting.

On a conceptual front, FL is viewed as being integral to privacy-preserving AI. Yang et al. (2019) have presented one of the seminal surveys conceptualizing adaptations of FL architectures for use cases extending from finance to healthcare. It is embodied in their observations, reminding one that FL is not diabetes-related, but its paradigm shifts are transferable to diabetes prediction at a global scale.

Overall, federated learning shows that high-quality AI models for diabetes prediction can be developed on a large scale in collaboration among different organizations while respecting data regulation constraints, thereby opening new pathways for future diabetes prediction projects at a global level.

7.5 Challenges faced during implementation

Data quality and adequacy are intrinsic obstacles for applying AI models for diabetes prediction because any errors or incomplete information presented within Electronic Health Records may adversely impact the performance of the AI model. Handling missing values or outliers is one of the primary strategies to cope with these obstacles (Wells et al. 2013). Another major limitation for AI is its acceptance among professionals, for which their involvement is required for AI development (Kelly et al. 2019).

Another significant challenge occurs because of the irregular or fluctuating natures of these processes and behaviors under real-world conditions, especially for settings where resources are limited and thus may have very different monitoring conditions. A key aspect of robust machine learning for such tasks is being able to learn under distributional shift techniques like domain adaptation to generalize robustly to naturally occurring variabilities under actual practice scenarios (Chen et al. 2019). The moral implications of data privacy concerns, bias within algorithms, and regulation requirements for GDPR and HIPAA guidelines make bias testing and large-scale data protection strategies imperative (Obermeyer et al. 2019). Finally, to implement AI within established healthcare systems, effective information technology collaboration for addressing technology-related challenges is necessary (Benis et al. 2021).

7.6 Influence on patient outcomes, healthcare expenses, and clinical decision-making processes

The use of AI models for diabetes prediction is thus very effective for patient care while at the same time ensuring low overall healthcare expenditure. Antecedent screening for patients at high risk helps to take preventive measures against diabetes and its complications. For instance, one study conducted among UK patients showed machine learning to have been very successful using EHR data to predict outcomes associated with diabetes and cardiovascular complications compared to traditional risk scores (Weng et al. 2017).

Also, AI models have helped to manage expenditure within the healthcare sector by preventing avoidable hospitalizations and costly treatments, thus offering cost advantages as well. A real-life study conducted by Dagliati et al. (2018) demonstrated the use of machine learning to correctly predict diabetes-related complications, thus preventing costly outcomes.

Additionally, AI-driven clinical decision support systems benefit providers by enabling them to adopt more focused and personalized approaches to treatment. Notably, Contreras and Vehi (2018) conducted a review on the use of AI in diabetes care and focused on its role in improving clinician confidence and patient engagement through decision support systems. Likewise, Tuppad and Patil (2022) conducted a systematic review on ML-driven decision support systems to highlight their role in error reduction and improved overall care.

In conclusion, it is apparent that for diabetes prediction and management, applying AI technology is effective in ensuring better patient outcomes and optimized efficiency of clinical decision-making while lowering healthcare expenses. Additionally, contemporary studies have also promoted the viability of applying multimodal and lightweight models for real-world applications. For instance, SACDNet (Nasir and Malik 2023) illustrated how uncertainty learning and attention learning can improve predictive outcomes using EHRs, while multimodal fusion for diabetes prediction is discussed by Ding et al. (2024).

7.7 Clinical workflow integration in resource-limited settings

AI deployment must align with existing care pathways to be useful, particularly where infrastructure and staffing are constrained. We outline a pragmatic workflow: (i) Screening at registration or triage using a short risk form (age, BMI, family history) on an offline-capable mobile app; (ii) Risk stratification with on-device or lightweight server inference (e.g., gradient-boosted trees), returning a simple Low/Moderate/High label plus two to three top SHAP-attributed factors; (iii) Action prompts mapped to local protocols (capillary glucose test, HbA1c order, lifestyle counseling, or referral); (iv) Follow-up via SMS/WhatsApp reminders and community-health-worker visits; and (v) Audit & feedback dashboards tracking positivity rate, adherence, and outcomes. Implementation considerations include HL7 FHIR or CSV-based interchange for clinics without full EHRs; offline-first data capture with secure store-and-forward; human-in-the-loop review for High-risk flags; alert governance (escalation limits, acknowledgment required); and equity checks (performance by age/sex/clinic). For physiological time-series (e.g., CGM or wearable data), we note that real-world signals often exhibit complex, partially coherent dynamics; therefore, we recommend robustness testing under stochastic perturbations, inspired by recent work on chaotic/coherent structure analysis in complex systems, which can guide stress-tests of model stability to signal irregularities.

7.8 Barriers to clinical adoption

Despite the great promise of AI-based diabetes prediction, several areas of challenge for full-scale clinical adoption remain. One is trust between physicians and AI algorithms because physicians remain unwilling to trust “black-box” algorithms lacking interpretability and training to make AI-driven decisions. Trust establishment among physicians also involves making these predictive outcomes interpretable while ensuring clinical validity and conformity to established guidelines for care. Another big challenge is EHR compatibility because most EHR systems remain substandard or proprietary or lack standard formats for data transfer among various systems to facilitate integration of predictive software into care pathways to improve efficiency instead of slowing down processes. Medico-legal concerns also remain big because most medico-legal liability for poor outcomes of AI-driven care is uncertain due to a lack of readiness to deal with medico-legal complications associated with these new types of care modalities, among others. Addressing these barriers through explainable AI, standardized data exchange protocols such as HL7 FHIR, clinician engagement, and supportive policies will be essential to ensure that AI models progress from promising prototypes to trusted clinical tools.

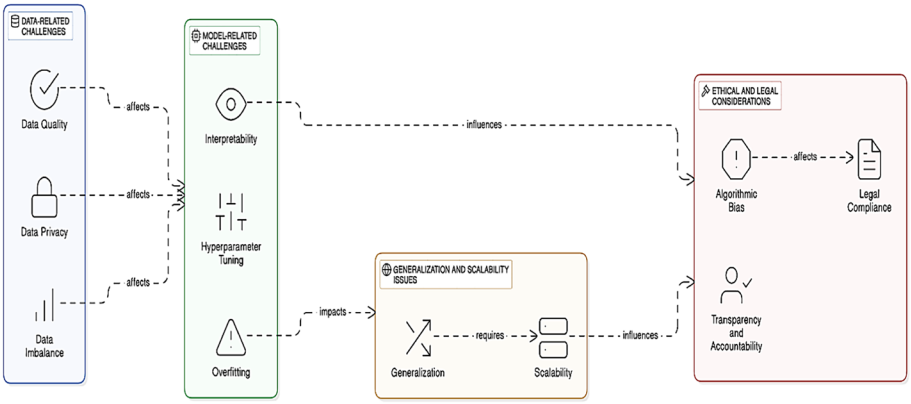

Fig. 12 AI model development challenges: addressing data, model, scalability, and ethical considerations for responsible AI deployment

8 Challenges and limitations

The issues discussed are relevant to the application of AI in diabetes prediction. To resolve the proper application of AI in the clinical environment, AI systems have to be properly and ethically designed, developed, and deployed. Generalization issues, data issues, model issues, and scalability issues are the most significant among them. All these issues need to be resolved to develop AI systems that are sustainable and trustworthy to implement in clinical environments.

Figure 12 shows data problems in models, such as scaling and ethics, as major obstacles to AI advancement. The effect of data problems on model explainability and generalizability, along with overfitting, is a major contributor to generalization and scalability. Ethical problems, such as transparency and bias, are a product of these data problems, thus affecting legal and ethical compliance.

8.1 Data-related challenges

Data quality issues are one of the most significant challenges in AI-driven diabetes prediction that compromise the accuracy and reliability of the model. One of the most obvious problems is data privacy since medical data is extremely sensitive and heavily protected by regulations like HIPAA in the US or GDPR in the EU. These regulations govern how patient data has to be stored, transmitted, and maintained so that the breach of confidentiality is never achieved. For this reason, it becomes very challenging to compile large datasets. Solutions for such conflicts have been suggested through data anonymization and federated learning at the cost of reduced utility of the data and compromised model accuracy (Lee et al. 2020). Class imbalance is another significant problem, where diabetic cases are very few as opposed to non-diabetic cases, and hence the predictions are biased and the model performance is affected. This leads to low-quality models that generalize well for the majority class but do not predict the minority class, leading to biased and unreliable results. With the help of oversampling methods like SMOTE, under-sampling, and even synthesizing data using GANs, these are some of the commonly used approaches that have tried to solve this

problem. These methodologies face shortcomings such as overfitting and the creation of fictional data instances . Models are also affected by the presence of quality issues, including missing values, inconsistencies, and forms of data noise. To solve these problems, the application of robust preprocessing methods, such as imputation and outlier detection, is required (Mohsen et al. 2023).

8.2 Model-related challenges

The algorithms employed by AI, and most importantly by computationally intensive endeavors such as DL, encounter several design-related problems. The most noteworthy is interpretability, especially given the health sector's reliance on models capable of giving understandable explanations and forecasts. However, advanced models such as deep neural models tend to have "black box" behavior, compromising clinicians' trust and confidence in their ability to predict.

Techniques associated with Explainable AI, including Shapley Additive Explanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME), have evolved to overcome such limitations; however, their effectiveness may be limited in the case of exceptionally complex models (Arrieta et al. 2020). The threat of overfitting is always a serious issue, where such models tend to become overly familiar with their training data, thus performing poorly on unseen and novel data. The phenomenon is observed to be particularly strong in DL models, which have a large ability to memorize their training data. To overcome the threat of overfitting, several strategies have been employed, including using regularizers (in particular, L1 and L2 regularizers), using cross-validation strategies, and pruning models.

Furthermore, while optimal performance through hyperparameter tuning is crucial, the procedure tends to be time-consuming and requires considerable computational power, especially while working on large datasets and computationally intensive models (Sudhakar et al. 2024).

8.3 Generalization and scalability considerations

To succeed in actual healthcare settings, AI models must generalize to new and unseen data and must be scalable to handle big datasets and real-time processing. Generalization is the major challenge since models learned from some datasets are not generalizable to other datasets from diverse populations due to demographic, genetic, and lifestyle variations. For instance, a model trained on data largely from a Caucasian population may exhibit low generalizability when evaluated on Asian or African populations. Transfer learning and domain adaptation techniques have been suggested to resolve this challenge; however, these also need validation on a variety of datasets. Recent surveys (e.g., Narasimharao et al. 2025) similarly stress that achieving generalizable and equitable AI systems remains a key challenge despite the rapid methodological progress in diabetes prediction.

Scalability is another important consideration, as the AI models would have to work on large data sets as well as in real time in actual healthcare settings. This would necessitate having computational infrastructure, like high-performance computers, to support this requirement. As more data is accumulated, the results of the AI models, which would not be updated, would tend to deteriorate in accuracy. For instance, federated learning, which

proposes to update AI models in multiple organizations without violating data confidentiality, has been suggested by authors such as Lee et al. (2020). However, issues such as communication costs and data heterogeneity (Rieke et al. 2020) also get raised in this scenario.

8.4 Ethical and legal issues

Even all pragmatic use of AI for diabetes detection in the healthcare industry is marred by legal and moral challenges of using AI itself. One of the most important moral challenges is algorithmic bias, which results in poor results for certain races or classes of people in society. Being in the healthcare industry, data is also regulated by certain legislation, for example, HIPAA, GDPR, to mention a few, meaning legal measures also require consideration from the compliance perspective. These regulations require that patient data is handled safely in addition to ensuring that the data is shared with, explained to, informed consent taken from, and understood by, the patient, who could be using it, meaning healthcare institutions have to provide clear explanations to patients regarding what they intend to do to them regarding data utilization (Wang et al. 2018). Multi-layered compliance is an altogether different story, as different measures, like different levels of privacy, multi-party computation, have been discussed to mitigate this challenge, but in this process, problems of poor performance of the model itself, as well as data usefulness, have also been raised (Vellido 2019). Morality also arises from a lack of transparency and accountability for this reason, because healthcare practitioners, as well as those who might get results from this data, for example, patient consideration, ought to be aware of predictive processes undertaken by AI in relation to results as well as who is to be blamed for adverse consequences, because data, which is skewed towards, for example, ethnic minorities, is used to train models, meaning that they end up failing people from other ethnicities as well, which is a recipe for increasing more of what is already common, obvious disparities in relation to healthcare (Wang et al. 2018). Furthermore, because of inconsistencies, ambiguity, and disparities brought about by AI, measures for clear actions and pragmatic policies require consideration (Johnson et al. 2019).

Apart from such general moral principles, fairness in AI systems needs to be evaluated systematically for subgroup results to prevent underserved communities from being adversely affected by AI systems. Some techniques for bias removal for AI systems include balanced data development for biased data sets for AI systems, adversarial debiasing techniques for biased data, and fairness-oriented learning techniques for biased data sets for AI systems, which have been adopted by current researchers for their AI studies for healthcare applications. Another point to consider for AI systems is regulation for compliance with strict guidelines for patient consent, data minimization, and audit trails for GDPR regulation in Europe and HIPAA guidelines for regulation in the United States of America. Despite challenges for scalable development of AI systems around different countries for diverse requirements and applications, they include specific guidelines for applying AI systems correctly and morally. Other advanced technology measures include differential privacy for regulation compliance for confidentiality of patient data, and meeting patient confidentiality concerns effectively and efficiently. Moreover, being quick remedies to problems of bias for AI algorithms and concerns for regulation and ethics for regulation and morality for AI systems are required for efficiency and effectiveness and for transparency for fairness and accountability for AI systems for healthcare applications, including an AI-based diabetes predictor system for healthcare for patients, which besides being faced with these points

also provides an unparalleled key to all healthcare systems for its effective influence on detection and efficient care for patients for better outcomes for patients.

8.5 Practical challenges in clinical deployment of AI

Though AI systems hold immense promise for diabetes forecasting, many challenges in real-world environments, apart from those of ethics, need to be tackled for implementing AI systems in healthcare settings (Poon et al., 2025). One of the biggest challenges is interfacing AI systems with present electronic health records (EHRs). Presently, many EHR systems have different data formats that make it difficult to implement a uniform interface, causing bottlenecks in healthcare by increasing costs for implementing this infrastructure.

Even in scenarios that allow for technological integration, trust and acceptance by medical professionals remain paramount for the realization of AI benefits. Professionals in the healthcare sector have aired grievances about "black box" systems, which function in closed systems, meaning that the functioning of those systems cannot be explained to confirm predictions made by those systems. A lack of transparency may lead to a loss of trust in the results of predictions by AI systems. Training of healthcare professionals is also paramount for a correct understanding of results from AI systems to aid in the responsible utilization of those results by healthcare professionals.

Obtaining regulatory approvals also makes it more complex to implement in a clinical setting. Standards such as HIPAA in the United States, for instance, as well as the EU's GDPR, set very stringent conditions for patient confidentiality, data protection, as well as patient consent in healthcare systems generally. Besides, obtaining regulatory approvals for using AI in healthcare applications is a complex, time-consuming procedure as well.

Operational constraints are more noticeable in resource-constrained environments, in which advanced AI algorithms, particularly deep learning, require considerable computational infrastructure for training as well as predictive work, which is an expensive requirement for most small, resource-constrained hospital settings in low- to middle-income countries. Even if feasibility is attained in this manner, costs associated with indigenous model training, as well as adequate support to ensure sustainability, may be prohibitively expensive in most instances, even in resource-constrained environments where hybrid strategies might alternatively prove more workable, although their efficacy also needs to be ascertained in this manner.

Finally, patient- as well as workflow issues also have to be taken into account to promote seamless integration of the systems. AI systems that require more data entry, impinge upon current workflows, or do not mesh well with current priorities of healthcare providers could end up slowing down the healthcare process rather than improving it.

In conclusion, in addition to pursuing issues of accuracy, fairness, and interpretability, achieving real-world effectiveness of AI for diabetes risk prediction in a healthcare environment also involves consideration of interoperability, physician acceptance, regulatory issues, infrastructure issues, and workflow integration as challenges to overcome to translate promising prototypes into sustainable, trustworthy tools for healthcare professionals.

8.6 Synthesis and roadmap

Taken together, these challenges can be addressed through targeted solutions and research directions. For example, data-related issues such as missing values, noise, and imbalance may be mitigated through robust preprocessing, synthetic data generation, and federated data-sharing frameworks. Model-related limitations, particularly interpretability and overfitting, are increasingly addressed by explainable AI methods (SHAP, LIME, attention) and regularization strategies. Challenges of generalization and scalability highlight the importance of domain adaptation, transfer learning, and the development of lightweight architectures for deployment in resource-constrained environments. Finally, ethical and regulatory concerns require the integration of fairness auditing, privacy-preserving frameworks (federated learning, blockchain), and compliance with standards such as GDPR and HIPAA. By linking each category of challenge with feasible solutions, this review identifies a roadmap for future work to make AI-driven diabetes prediction more robust, equitable, and clinically actionable. In parallel, conversational AI and large language models (LLMs) are beginning to reshape the healthcare AI landscape. Recent work by (Masood et al. 2023) outlines their potential roles in diagnostics, clinical decision support, and patient engagement. While distinct from predictive modeling, these generative approaches complement diabetes prediction by broadening the ecosystem of AI-driven healthcare tools, underscoring the need for integration across multiple AI paradigms.

9 Conclusion and future directions

AI innovations hold enormous potential in improving diabetes diagnosis, targeted intervention, and outcomes, but to actualize this potential, emphasis needs to be placed on several critical areas. First, work on lightweight transformer models would prove critical for real-world applications in resource-limited healthcare settings. Second, development work in privacy-preserving methods, such as federated learning using blockchain and secure multi-party computation, needs to proceed to ensure confidentiality of patient data during collaborative modeling. Third, data integration methods using multiple data types, including electronic health records, wearable data, imaging data, and genomic data, promise immense opportunities for precision in diabetes modeling. Finally, combining AI methods, such as SHAP, LIME, and attention, would prove pivotal for clinician trust-building, regulatory approvals, and successful diabetes modeling applications in healthcare settings. While this analysis has several potential drawbacks, including variations in study protocols, use of only normative datasets characterized by homogeneous populations, and time-related obsoleteness of rapidly shifting technologies, including AI, to an extent, this review provides a coherent perspective into AI-based promising diabetes models, pointing to fundamental priorities for progressing from promising technologies to efficient applications in diabetes prediction modeling of real-world interest.

One of the most apparent results of this review is that it is difficult to identify an absolute best-performing model for all AI; this is greatly dependent upon data modality. While RF/XGBoost is best for structured EHRs, CNN is best for instances of imaging, such as those for eye diagnosis in retinal scans, whereas LSTM and transformer models shine in cases of time-series data from CGM systems and EHRs, respectively. While understanding this

modality-specific performance is key to identifying whether an AI model is useful, it is also critical to understand what makes a model useful in different contexts.

Overall, this review provides a unified methodological and clinical synthesis of AI-driven diabetes prediction, bridging diverse data modalities, EHRs, imaging, wearables, and continuous glucose monitoring, with emerging paradigms such as federated learning and explainable AI. By aligning technical innovation with clinical applicability, the paper offers a comprehensive roadmap for advancing transparent, generalizable, and practice-ready AI systems in diabetes care.

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Author contributions Saleha Masood (Corresponding Author): Led the conception and design of the study, formulated the review protocol, conducted an extensive literature search and screening, analyzed and synthesized the data, and drafted the initial and revised versions of the manuscript. Coordinated contributions from all co-authors and managed the submission and revision process. Mousa Ahmad Al-Bashrawi: Contributed to the development of the research framework and evaluation criteria for included studies, supported the literature screening and data extraction process, and offered critical insights into the integration of AI techniques within healthcare systems. Assisted in the interpretation of results from both a technical and clinical perspective, and helped refine the manuscript for clarity and coherence. Muhammad Attique Khan: Provided specialized expertise on ML and DL algorithms, particularly in the context of biomedical data. Assisted in evaluating technical strengths and limitations of reviewed models, contributed to the analysis of performance metrics, and helped shape the discussion on emerging AI approaches such as federated learning and explainable AI. Yogesh K. Dwivedi: Guided the strategic direction of the review and its alignment with ongoing research trends in digital health and AI applications. Contributed to refining the theoretical framework and ensuring methodological rigor. Provided critical feedback on all sections of the manuscript, enhanced the narrative flow, and supported the articulation of implications for researchers, practitioners, and policymakers. All authors reviewed and approved the final version of the manuscript.

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Declarations

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